Elimination of the GGHV Branch-(a,b) Normal Form at Degree Pair $(72, 108)$:

A Certified Computational Audit

Layer Reduction, Top-Layer Classification, and Final Obstruction

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Computation, Lean and LaTeX Source:
https://github.com/brandonmccraryresearch-cloud/2D-Jacobian-Conjecture-Workspace

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Abstract

We present a computational elimination of the branch (a,b) normal form arising from Proposition 4.3 of Guccione–Guccione–Horruitiner–Valqui [1] for a hypothetical $(72, 108)$ counterexample to the planar Jacobian conjecture. Every base object is source-gated directly against GGHV v1 before any proof is drafted.

Starting from the verified Newton polygons

$$N(P) = \mathrm{conv}\{(0,0),(1,0),(8,14),(8,16)\},$$
$$N(Q) = \mathrm{conv}\{(0,0),(2,1),(12,21),(12,24)\},$$

with $[P, Q] = x^2$ post-φ, we reduce the 72-unknown bilinear system to five one-variable layers $E_1, \dots, E_5$ totaling 92 equations, verified as an exact symbolic polynomial identity.

The top layer E_5 is the Davenport–Stothers extremal equation $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ [2, 3]. The Belyi classification gives the geometric upper bounds 3, 10, 35 chart points for $m = 3, 5, 7$; for $m = 7$ the bound is sharp, with the degree-35 eliminating polynomial $W(a_6)$ irreducible over $\mathbb{Q}$, a quintic in $S = a_6^7$ with nonzero discriminant and exactly one real root. The 35 solutions are distinct and form a single Galois orbit: one real solution and 17 conjugate pairs, grouped in five S-orbits. The classification is proved twice: the Belyi bound combined with the 35 exhibited points gives a complete classification (relying on Riemann's existence theorem and the computed msolve parametrization), while §6.4 gives an independent algebraic proof, valid over every field of characteristic 0, which is the route formalized in Lean. In reversed coordinates the chart equations are the coefficient recursion of $(1 + y_1 v + \cdots + y_7 v^7)^{3/2}$. Six weighted-homogeneous relations, of weights $4, 5, 5, 6, 6, 7$ and with explicit certificates, cut out the five torus orbits. We formalize this proof in Lean.

For $m = 3$ and $m = 5$ the bounds 3 and 10 are also sharp. These two classifications are kernel-checked in Lean as well, over every field of characteristic 0. In the normalized chart, the system holds if and only if two things hold: the eliminant vanishes ($3a_2^3 - 32$ for $m = 3$, $9a_4^{10} + 37200a_4^5 + 95051008$ for $m = 5$), and the remaining unknowns take explicit values. In each case the eliminant has no repeated root. So over every algebraically closed field of characteristic 0 there are exactly 3 solutions for $m = 3$ and 10 for $m = 5$, and this count is kernel-checked too. Both eliminants are irreducible over $\mathbb{Q}$, and this is kernel-checked as well. Theorem 1.1 does not need these two cases.

The descent proceeds through E_4 (an 18×19 matrix of rank 17, kernel dimension 2), E_3 (a 19×20 matrix of rank 18, kernel dimension 2, unconditionally solvable for every direction t), and E_2 (injective 19×12, rank 12), leaving seven explicit compatibility conditions. We expand these as $b_i(t) + s_1 M_i(t) + s_2 L_i(t) = 0$ with b_i cubic and M_i, L_i linear, eliminate (s_1, s_2) to obtain 35 quintic 3×3 minors. The entire descent is computed exactly over the quintic field K_5 in characteristic 0: the 35 minors span all binary quintics, forcing $t = 0$. The case $t = 0$ forces B_0 constant, contradicting the required vertex $(12, 24)$. The Lean formalization quantifies over all roots of R and all torus parameters, covering all 35 top-layer solutions.

Hence no polynomials P, Q satisfying NewtonNF2 have $[P, Q] = \lambda x^2$ with $\lambda \neq 0$ (Theorem 1.1). In Lean this NewtonNF2 form is a single kernel-checked theorem, `main_theorem`, on the standard axioms only, independent of GGHV Proposition 4.3. Its top-layer input is `chartClassification_holds`, a formalized algebraic proof of the $m = 7$ top-layer classification of Proposition 6.1, in the normalized chart (§6.4). The further rigidity $a_{8,16} = 0$ (Corollary 1.2) is kernel-checked in Lean as well, together with the vanishing of every other coefficient of P and Q off the top edges except the constant terms (`lower_edge_rigidity`); it uses the layer E_1. A second proof of $a_{8,16} = 0$, outside Lean and independent of the descent, is an explicit certificate over K_5: the identity $a_{8,16}^2 = \sum_{k=1}^{75} H_k e_k$, where the e_k are the 75 equations of the layers $E_4, \dots, E_1$. Two independent programs check it exactly. The application to the Jacobian conjecture is conditional on the correctness and exhaustiveness of [1, Proposition 4.3] (Corollary 1.4).

Contents

1 Introduction **4**

1.1 Context and prior work 4

1.2 Main result 4

1.3 Logical status 5

1.4 Outline 6

2 Source data **7**

3 The polynomial system **7**

3.1 Lattice points 7

3.2 The bracket equations 8

4 Exact layer reduction **8**

5 Normalization **10**

6 The top layer **10**

6.1 The Davenport–Stothers equation . . . 10
6.2 Classification . . . 11
6.3 Irreducibility and reality . . . 18
6.4 An algebraic proof of the top-layer classification for $m = 7$, and its formalization 18

7 Descent **23**
7.1 Layer E_4 . . . 23
7.2 Layer E_3 . . . 23
7.3 Layer E_2 . . . 24

8 The final obstruction **24**
8.1 Explicit form of the seven conditions . . . 25
8.2 Elimination of (s_1, s_2) . . . 25
8.3 The case $t = 0$ . . . 26
8.4 Proof of the main theorem . . . 26

9 Exact characteristic-0 descent **29**

10 Conclusion **30**

11 Prior work, attribution, and independence **31**

A Computational archive **32**

B Original certificate (second route) **33**

C Computational and AI-assisted research disclosure **34**

1 Introduction

1.1 Context and prior work

The planar Jacobian conjecture asserts that a polynomial map $(P, Q) : \mathbb{C}^2 \to \mathbb{C}^2$ with $[P, Q] := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ is invertible. In [1], Guccione, Guccione, Horruitiner, and Valqui raise the lower bound on $\max\{\deg P, \deg Q\}$ for a hypothetical counterexample from 100 (Moh [4]) to 108, leaving open only the degree pair $(72, 108)$ and its symmetric $(108, 72)$.

Santibáñez-Leal has published machine evidence related to the $(72, 108)$ candidate [5, 6], explicitly not claiming elimination. The candidate itself is documented in [7].

For the degree pair $(72, 108)$, [1] (an unrefereed preprint, arXiv:2204.14178v1) identifies two cases. The corner $(9, 27)$ is discarded in [1, §5]; the open case is the corner $(8, 28)$ with $(m, n) = (3, 2)$. For this case, [1, Proposition 4.3] shows that any counterexample gives rise, after an explicit (non-invertible) morphism φ, to one of two Newton polygon configurations. Case (1), with additional vertices $(0, 8)$ resp. $(0, 12)$, is branch (c). Case (2), the subject of this paper, is branches (a) and (b):

$$\begin{aligned} N(P) &= \operatorname{conv}\{(0,0), (1,0), (8,14), (8,16)\}, \\ N(Q) &= \operatorname{conv}\{(0,0), (2,1), (12,21), (12,24)\}, \end{aligned} \tag{1}$$

with $[P, Q] = x^2$ in post-φ coordinates. See Figure 1 for the polygon shapes.

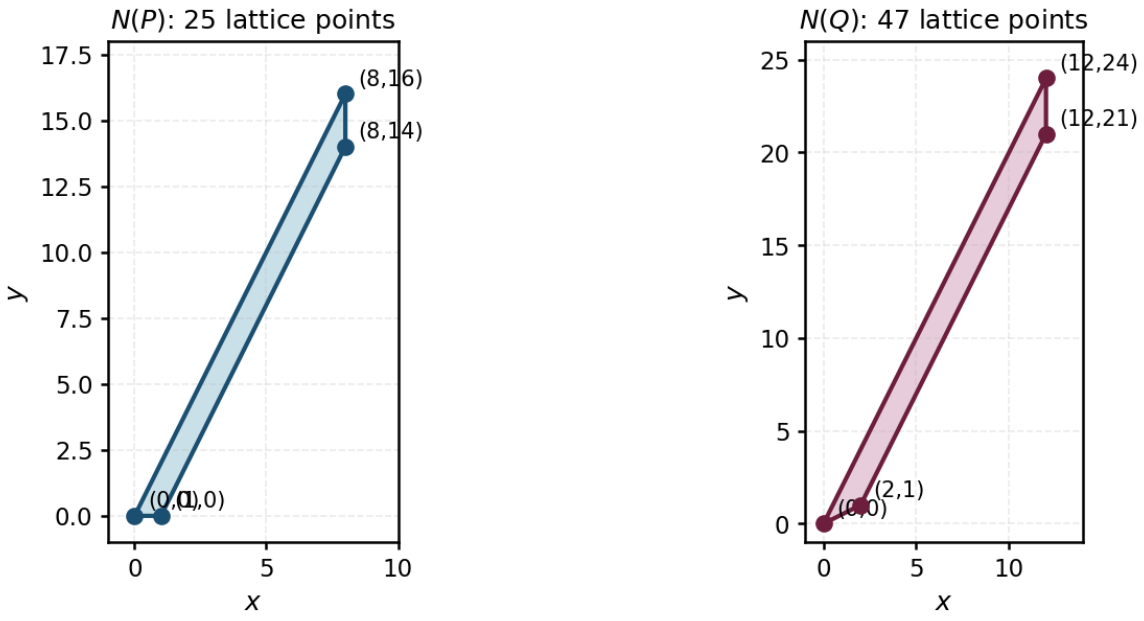


Figure 1: The branch (a,b) Newton polygons from [1, Proposition 4.3(2)], source-gated at v1 lines 495 and 599–600. $N(P)$ has 25 lattice points; $N(Q)$ has 47.

1.2 Main result

Theorem 1.1 (Elimination of the GGHV branch-(a,b) normal form)**.** *Let K be a field of characteristic 0. Let $P, Q \in K[x, y]$ have* $\operatorname{supp} P \subset N_P$ *and* $\operatorname{supp} Q \subset N_Q$*, with $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot$*

$b_{12,21} \neq 0$ *and* $[P,Q] = \lambda x^2$ *for some* $\lambda \neq 0$*. Then* $b_{12,24} = 0$*. In particular, no* P, Q *satisfy* NewtonNF2(P,Q) *(listing 11; the polygons are in §2) with* $[P,Q] = \lambda x^2$*.*

Corollary 1.2 (Lower-edge rigidity)**.** *Under the hypotheses of Theorem 1.1,* $a_{8,16} = 0$ *as well. More precisely, every coefficient* $a_{i,j}$ *with* $j \geq 2i - 1$ *and every coefficient* $b_{i,j}$ *with* $j \geq 2i - 2$ *vanishes, except* $a_{0,0}$ *and* $b_{0,0}$*:* $P = P_2 + a_{0,0}$ *and* $Q = Q_3 + b_{0,0}$*. In Lean this is* ***`lower_edge_rigidity`*** *(listing 21), kernel-checked on the standard axioms;* $\lambda \neq 0$ *is not needed. A second proof of* $a_{8,16} = 0$*, outside Lean, is an explicit certificate over* K_5 *(****`scripts/a816_certificate/`****), checked exactly by* Singular *and by python-flint; see §5. The corollary is not needed for the elimination:* $b_{12,24} = 0$ *already contradicts the required Newton vertex* $(12, 24)$*.*

Remark 1.3 (Lean-to-mathematics correspondence)*.* The Lean predicate `NewtonNF2` (listing 11) is

$$\texttt{NewtonNF2}(P,Q) \iff \begin{cases} \operatorname{supp} P \subseteq N_P, \ \operatorname{supp} Q \subseteq N_Q \text{ (via } \texttt{inNP}, \texttt{inNQ}), \\ a_{0,0}, a_{1,0}, a_{8,14}, a_{8,16} \neq 0, \\ b_{0,0}, b_{2,1}, b_{12,21}, b_{12,24} \neq 0, \end{cases}$$

i.e. exactly the support conditions of Theorem 1.1 together with the nonvanishing of the eight vertex coefficients. The half-plane predicates are tied to the vertex sets by `decide` checks run on duplicate predicates `latticeNP`/`latticeNQ` (which restate the same inequalities): the lattice-point counts (25 and 47) and the vertex-membership check confirm the eight vertices lie on the polygons. This bridge is what lets the kernel-checked `main_theorem` speak about the same P, Q as Theorem 1.1.

Corollary 1.4 (Conditional on GGHV)**.** *Assuming [1, Prop. 4.3], no counterexample to the plane Jacobian conjecture arises in case* $(8, 28)$*, sub-cases a) and b).*

The proof is computational but exact. We reduce the 72-unknown system to triangular layers (Figure 2), determine the top layer via the Belyi-map/Riemann-existence argument (Figure 3), and independently by an algebraic argument formalized in Lean (§6.4). We then obstruct the descent via the Lean-formalized layer certificates (§7). Key steps were verified in Singular 4.4.1 and Python/SymPy; scripts are archived with this draft.

1.3 Logical status

Remark 1.5. Theorem 1.1 does not depend on GGHV. Its proof uses Proposition 6.1 for $m = 7$, which is proved twice. On paper it follows from Riemann's existence theorem (§6). In §6.4 the classification it provides is proved algebraically, in the normalized chart, and that proof is formalized in Lean as `chartClassification_holds` (listing 16). In Lean, `main_theorem_of_chart` (listing 7) derives the NewtonNF2 form of Theorem 1.1 from the explicit statement `ChartClassification` (listing 12). `main_theorem` (listing 17) combines it with `chartClassification_holds` into a single kernel-checked theorem independent of GGHV Proposition 4.3, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The $b_{12,24} = 0$ part follows from `no_completion_K5_PQ` together with `topLayerClassification_of_chart` and `chartClassification_holds`; the $a_{8,16} = 0$ part (Corollary 1.2) is `a816_eq_zero`, and the whole corollary is `lower_edge_rigidity` (listing 21); both are kernel-checked on the same three axioms and reuse the descent up to $t = 0$. Outside Lean, an explicit certificate $a_{8,16}^2 = \sum_k H_k e_k$ over K_5, stored as the 76 cofactors of $1 = \sum_k z^2 H_k e_k - (1 + a_{8,16} z)(a_{8,16} z - 1)$ in `scripts/a816_certificate/a816_lift.txt` and checked exactly by Singular and by python-flint, proves $a_{8,16} = 0$ without the descent (see §5). Only Corollary 1.4 also assumes GGHV Proposition 4.3.

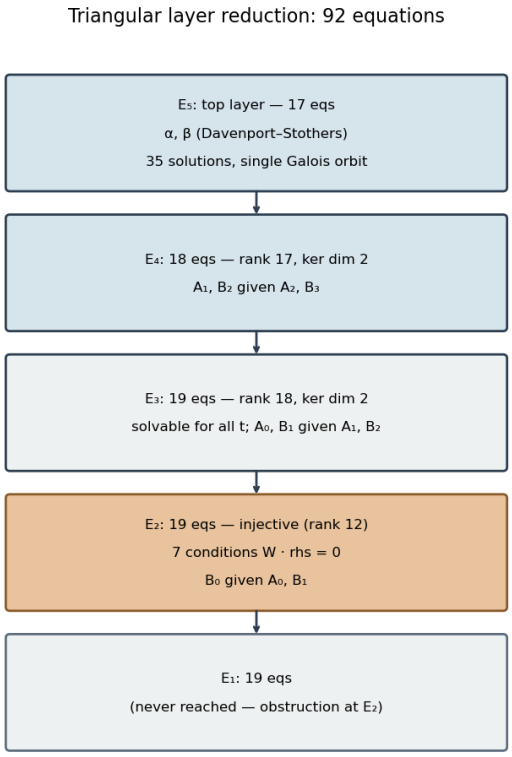


Figure 2: The triangular layer structure: 92 equations in five layers. Each layer is affine-linear in its unknowns given the solution above it.

1.4 Outline

Section 2 source-gates all base objects against GGHV v1. Section 3 sets up the 72-unknown system. Section 4 gives the exact layer reduction. Section 5 fixes normalizations. Section 6 classifies the top layer. Section 7 carries out the descent. Section 8 proves the final obstruction. Section 9 lifts to characteristic zero. Section 10 concludes and discusses the remaining branch (c). Section 11 covers prior work, attribution, and independence.

The proof is organized around four certificates. *Certificate I* (Newton reduction): $[P,Q] = \lambda x^2 \Rightarrow E_5, \dots, E_1$ (§4). *Certificate II* (top-layer completeness): $E_5 \Rightarrow$ five torus orbits over K_5 (§6, §6.4). *Certificate III* (descent obstruction): $E_4, E_3, E_2 \Rightarrow$ the compatibility ideal equals $(t_1, t_2)^5$, hence $t = 0$ (§7, §8). *Certificate IV* (vertex destruction): $t = 0 \Rightarrow B_0' = 0 \Rightarrow b_{12,24} = 0$ (Corollary 8.6). The rigidity $a_{8,16} = 0$ (Corollary 1.2) is a supplementary result; Theorem 1.1 does not use it. The dependency graph is

$$\text{Counterexample in case } (8,28) \xrightarrow{\text{GGHV 4.3}} \text{NewtonNF2}, [P,Q] = \lambda x^2 \xrightarrow{\text{Thm. 1.1}} \text{contradiction}$$

with Theorem 1.1 itself independent of GGHV:

Result	Depends on GGHV?
Layer reduction	No
Top-layer classification	No
K_5 descent	No
$b_{12,24} = 0$	No
$a_{8,16} = 0$	No
Theorem 1.1	No
Corollary 1.4 (Jacobian application)	Yes

2 Source data

We verify every base object directly against version 1 of [1] (arXiv:2204.14178v1), downloaded from `https://arxiv.org/pdf/2204.14178v1` and extracted with `pdftotext -layout`. No proof is drafted until the polygons, bracket, and φ are confirmed character-by-character against GGHV's text.

Proposition 2.1 ([1, Proposition 4.3], source-gated)**.** *If there is a counterexample to the Jacobian conjecture in case* $(8, 28)$*, then there exist* $P, Q \in L^{(1)}$ *with* $[P, Q] = x^2$ *and one of:*

(1) $N(P) = \{(0,0), (1,0), (8,14), (8,16), (0,8)\}$,
$N(Q) = \{(0,0), (2,1), (12,21), (12,24), (0,12)\}$;
(2) $N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$,
$N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$.

Source locations. The `pdftotext -layout` extraction of v1 yields:

- Line 33: $[P, Q] := P_x Q_y - P_y Q_x \in K^\times$ (bracket convention).
- Line 492: "Proposition 4.3 (Case $(8, 28)$)."
- Line 493: "with $[P, Q] = x^2$" (rendered as `x2` in the extraction).
- Line 495: Case (2) polygons
$N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$, $N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$.
- Lines 596–597: The morphism $\varphi(x) = x^{-1}$, $\varphi(y) = x^4 y$; "this is not an automorphism and the chain rule gives $[\varphi(P), \varphi(Q)] = -[P, Q]x^2$."
- Lines 599–600: "in the cases a) and b) the Newton Polygons of P and Q become $N(P) = \{(0,0), (1,0), (8,14), (8,16)\}$, $N(Q) = \{(0,0), (2,1), (12,21), (12,24)\}$."

A verification script (`source_gate_verify.py`) asserts each of these lines automatically; it exits non-zero if any check fails. □

We work exclusively with case (2) (branches a,b).

3 The polynomial system

3.1 Lattice points

The Newton polygons (1) contain:

$$|N(P) \cap \mathbb{Z}^2| = 25, \qquad |N(Q) \cap \mathbb{Z}^2| = 47, \tag{2}$$

verified by direct enumeration (cross-checked with Pick's theorem: area$(N(P)) = 15$, boundary 18, interior 7). This gives $25 + 47 = 72$ unknown coefficients. The two constant terms do not appear in $[P, Q]$, leaving 70 effective unknowns, but we retain all 72 for the layer structure.

3.2 The bracket equations

Writing $P = \sum_{(i,j)\in N(P)} a_{ij} x^i y^j$ and $Q = \sum_{(i,j)\in N(Q)} b_{ij} x^i y^j$, the condition $[P, Q] = x^2$ expands to bilinear equations in the coefficients. Solving the full 72-unknown bilinear system directly is impractical; we use the exact layer reduction of §4. (Direct expansion remains feasible for *verification* of the reduction, as in Proposition 4.1.)

4 Exact layer reduction

Set $u = xy^2$ and decompose into y-layers:

$$P = \sum_k y^{-k} A_k(u), \qquad Q = \sum_\ell y^{-\ell} B_\ell(u), \tag{3}$$

where $A_k, B_\ell \in \mathbb{C}[u]$ are one-variable polynomials whose coefficients are linear forms in the a_{ij}, b_{ij}. The bracket $[P, Q]$ becomes a sum over y-layers, each a one-variable identity in u.

Proposition 4.1 (Exact layer reduction)**.** *The system $[P, Q] = x^2$ on the 72 coefficients implies five one-variable polynomial identities $E_1, \dots, E_5$ (the converse is not needed):*

Layer	*Equations*	*Unknowns (given previous layers)*
E_5	17	*Top:* α, β
E_4	18	A_1, B_2 *given* A_2, B_3
E_3	19	A_0, B_1 *given* A_1, B_2
E_2	19	B_0 *given* A_0, B_1
E_1	19	*Lowest coefficients*
Total	92	

The layer reduction was verified as a polynomial identity (script ***C1_layer_reduction.sing****) with all 72 coefficients treated as symbolic indeterminates; the residual is identically zero.*

Computational verification. Substituting (3) into $[P, Q] - x^2$ and collecting by powers of y yields the five identities $E_1, \dots, E_5$. Expanding each E_i in the 72 indeterminates and comparing coefficients with the direct bilinear expansion of $[P, Q] - x^2$ gives zero difference as a polynomial in 72 variables. Script: `C1_layer_reduction.sing` (SINGULAR). □

Listing 1: Mechanically extracted from ***Jacobian/BranchAbLayers.lean*** *(sha256* ***7c16141b4b126b3e****); statement verbatim, proof elided.*

```
theorem layers_of_jac (lam : K) (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : Polynomial K)
    (p₀ p₁ p₂ q₀ q₁ q₂ q₃ : MvPolynomial (Fin 2) K)
    (hp₀ : X 1 ^ 0 * p₀ = evH A₀) (hp₁ : X 1 ^ 1 * p₁ = evH A₁) (hp₂ : X 1 ^ 2 * p₂ =
     ↪  evH A₂)
```

```
    (hq₀ : X 1 ^ 0 * q₀ = evH B₀) (hq₁ : X 1 ^ 1 * q₁ = evH B₁) (hq₂ : X 1 ^ 2 * q₂ =
     ↪ evH B₂)
    (hq₃ : X 1 ^ 3 * q₃ = evH B₃)
    (hJ : jac (p₀ + p₁ + p₂) (q₀ + q₁ + q₂ + q₃) = C lam * X 0 ^ 2) :
    layerTerm 2 3 A₂ B₃ = Polynomial.C lam * Polynomial.X ^ 2 /\
    layerTerm 2 2 A₂ B₂ + layerTerm 1 3 A₁ B₃ = 0 /\
    layerTerm 2 1 A₂ B₁ + layerTerm 1 2 A₁ B₂ + layerTerm 0 3 A₀ B₃ = 0 /\
    layerTerm 2 0 A₂ B₀ + layerTerm 1 1 A₁ B₁ + layerTerm 0 2 A₀ B₂ = 0 /\
    layerTerm 1 0 A₁ B₀ + layerTerm 0 1 A₀ B₁ = 0 := by
  -- proof body elided (see artifact)
```

The basic definitions used above are printed verbatim here, since only a human reader (not the Lean kernel) checks definitions:

*Listing 2: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def jac (P Q : MvPolynomial (Fin 2) K) : MvPolynomial (Fin 2) K :=
  pderiv 0 P * pderiv 1 Q - pderiv 1 P * pderiv 0 Q
```

*Listing 3: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def uu : MvPolynomial (Fin 2) K := X 0 * X 1 ^ 2
```

*Listing 4: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def evH : Polynomial K →ₐ[K] MvPolynomial (Fin 2) K := Polynomial.aeval (uu (K := K))
```

*Listing 5: Mechanically extracted from **Jacobian/BranchAbLayers.lean** (sha256 **7c16141b4b126b3e**); definition printed verbatim in full.*

```
def layerTerm (a b : Nat) (A B : Polynomial K) : Polynomial K :=
  (a : Polynomial K) * A * Polynomial.derivative B - (b : Polynomial K) *
   ↪ Polynomial.derivative A * B
```

The system is triangular (Figure 2): E_5 involves only the top-layer unknowns, and each E_i ($i < 5$) is affine-linear in its unknowns given a solution of the layers above. This is the key structural fact that makes the problem tractable. Conceptually, the Newton polygons induce a weighted grading, and the Jacobian equation respects that grading: the five E_i are the homogeneous components of $[P, Q] = x^2$ with respect to it. The triangular structure is thus not a computational accident but a structural consequence of the grading, and the operator $\mathrm{LT}_{a,b}(A, B) = aAB' - bA'B$ is its homogeneous bracket.

5 Normalization

The top-layer unknowns admit rescaling symmetries from the torus action. We fix, without loss of generality,

$$a_{1,0} = a_{8,14} = 1, \tag{4}$$

leaving $a_{8,16}$ free. The remaining 1-dimensional torus acts by the Y-weight grading used in the layer decomposition (§4); the finite subgroup μ_7 gives 7 chart copies per orbit.

Remark 5.1 (Lean chart normalization). The Lean formalization uses chart coordinates $(a_1, \dots, a_7, b_0, \dots, b_9)$ with the torus normalization $a_7^3 b_0^2 = 1$ (see `ChartClassification` in `Jacobian/BranchAbChart.lean`). After fixing $\alpha_0 = 1$ and $\beta_{10} = 1$, the remaining scaling $u \mapsto \kappa u$ acts on a_7 with weight 7 and on b_0 with weight -10, hence on $a_7^3 b_0^2$ with weight $21 - 20 = 1$. The condition $a_7^3 b_0^2 = 1$ therefore fixes κ completely, with no 7th-root ambiguity: the Lean chart meets each torus orbit *exactly once*, giving 5 points, not 35. The §5 chart ($a_{1,0} = a_{8,14} = 1$) meets each orbit 7 times. The two charts are related by a torus element requiring a 7th root.

Vertex conditions used. The proof uses $(1, 0)$ and $(2, 1)$ (equivalently $\lambda \neq 0$), $(8, 14)$ (normalization and top-layer degree), and $(12, 24)$ (the $t = 0$ exclusion). The vertex $(8, 16)$ is left free in `main_theorem`. Once E_1 is included, $a_{8,16} = 0$ is forced (Corollary 1.2). This gives a second contradiction, from the lower edge: `main_theorem_lower_edge` (listing 22) refutes NewtonNF2 without using the vertex $(12, 24)$. It shares the descent up to $t = 0$ with `main_theorem`. The explicit certificate over K_5 (`scripts/a816_certificate/`, checked exactly by SINGULAR and by python-flint) reaches $a_{8,16} = 0$ without the descent, outside Lean.

Listing 6: Mechanically extracted from ***`Jacobian/BranchAbTorus.lean`*** *(sha256* ***`97918673d5c4dc09`****); statement verbatim, proof elided.*

```
theorem layers_transport (α β κ : K) (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : K[X])
    (E4 : layerTerm 2 2 A₂ B₂ + layerTerm 1 3 A₁ B₃ = 0)
    (E3 : layerTerm 2 1 A₂ B₁ + layerTerm 1 2 A₁ B₂ + layerTerm 0 3 A₀ B₃ = 0)
    (E2 : layerTerm 2 0 A₂ B₀ + layerTerm 1 1 A₁ B₁ + layerTerm 0 2 A₀ B₂ = 0) :
    layerTerm 2 2 (C α * sc κ A₂) (C β * sc κ B₂) + layerTerm 1 3 (C α * sc κ A₁) (C
     ↪  β * sc κ B₃) = 0 /\
    layerTerm 2 1 (C α * sc κ A₂) (C β * sc κ B₁) + layerTerm 1 2 (C α * sc κ A₁) (C
     ↪  β * sc κ B₂) +
      layerTerm 0 3 (C α * sc κ A₀) (C β * sc κ B₃) = 0 /\
    layerTerm 2 0 (C α * sc κ A₂) (C β * sc κ B₀) + layerTerm 1 1 (C α * sc κ A₁) (C
     ↪  β * sc κ B₁) +
      layerTerm 0 2 (C α * sc κ A₀) (C β * sc κ B₂) = 0 := by
  -- proof body elided (see artifact)
```

6 The top layer

6.1 The Davenport–Stothers equation

The top identity E_5 takes the form

$$\alpha\beta + u\,(2\alpha\beta' - 3\alpha'\beta) = 1, \tag{5}$$

where $\alpha = A_2/u$ and $\beta = B_3/u^2$ are polynomials of degree 7 and 10 respectively, the degrees being forced by the vertices $(8, 14) \in N(P)$ and $(12, 21) \in N(Q)$.

Setting $u = s^2$ with $f(s) = \alpha(s^2)$ and $g(s) = s\beta(s^2)$, equation (5) becomes the Davenport–Stothers extremal equation [2, 3]

$$2fg' - 3f'g = c \neq 0. \tag{6}$$

Indeed $2fg' - 3f'g = 2(\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta))$. With $\deg f = 14 = 2k$ and $\deg g = 21 = 3k$ ($k = 7$), we have $\deg(g^2 - c \cdot f^3) = 8 = k + 1$, which is exactly Davenport's bound: the pair is extremal. with f even and g odd. Here $\deg_s f = 2m$ where $m = \deg_u \alpha$; the relation $1 + 2n - 3m = 0$ (with $n = \deg_u \beta$) forces m odd. The cases $m = 1, 3, 5, 7$ admit solutions; $m = 7$ is the relevant case for (1). The symmetric (parity-constrained) classification is computed here via elimination; we do not claim it follows from the literature.

6.2 Classification

Proposition 6.1 (Top-layer solutions)**.** *The system E_5 has:*

- $m = 3$*: 3 complex solutions; eliminant* $3a_2^3 - 32$*, irreducible over* $\mathbb{Q}$*;*
- $m = 5$*: 10 complex solutions; eliminant* $9a_4^{10} + 37200a_4^5 + 95051008$*, irreducible over* $\mathbb{Q}$*;*
- $m = 7$*: 35 complex solutions; eliminant* $W(a_6)$ *of degree* 35*.*

The $m = 7$ case is the one relevant to (1)*.*

Proof. We work over $\mathbb{C}$: the Belyi classification is topological, hence valid over $\mathbb{C}$, and the resulting count of solutions transfers to any field of characteristic 0 by base change (the equations are defined over $\mathbb{Q}$). We normalize the third branch point to $c = 1$ by the scaling $u \mapsto cu$, which preserves the torus action. We prove the counts in characteristic 0 via the Belyi-map classification. For the top layer, E_5 is $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta) = 1$ with $\deg\alpha = m$, $\deg\beta = (3m - 1)/2$ ($m = 3, 5, 7$, so $\deg\beta = 4, 7, 10$). Put $\varphi = u\beta^2/\alpha^3$. Then $\varphi' = \beta E/\alpha^4 = \beta/\alpha^4$, so φ is a Belyi map: it ramifies only over $0, \infty$, and $c = \varphi(\infty) \neq 0$. The ramification is forced as follows. At a root r of α, $E(r) = -3r \cdot \alpha'(r)\beta(r) = 1$. So α has simple nonzero roots, none shared with β. At a root s of β, $E(s) = 2s \cdot \alpha(s)\beta'(s) = 1$, so β also has simple nonzero roots. Hence $\deg\varphi = 3m$, and $\varphi' = \beta/\alpha^4$ vanishes in $\mathbb{C}$ only at the roots of β, which forces the passport

$m = 3$	$[2^4 1]$ $(n = 9)$	$[3^3]$	$[7 \cdot 1^2]$
$m = 5$	$[2^7 1]$ $(n = 15)$	$[3^5]$	$[12 \cdot 1^3]$
$m = 7$	$[2^{10} 1]$ $(n = 21)$	$[3^7]$	$[17 \cdot 1^4]$

By Riemann's existence theorem, covers with this passport correspond to S_n-conjugacy classes of transitive triples in $C_0 \times C_1 \times C_\infty$ with product 1 (here $n = 3m$). The Frobenius character formula (computed exactly in `belyi_counts_m357.py` via the Murnaghan–Nakayama rule) counts N triples with $N = 9!, 2 \cdot 15!, 5 \cdot 21!$ for $m = 3, 5, 7$; since every such triple has trivial centralizer, each S_n-conjugacy class has $n!$ elements, giving $N/n! = 1, 2, 5$ covers. Every triple counted is transitive, for general m: a non-transitive triple would have an orbit on which g_∞ acts trivially, so on that orbit $g_0 = g_1^{-1}$; but g_1 has cycle type 3^m (hence fixes no point, and so does g_1^{-1}), while g_0 has cycle type $2^{(3m-1)/2}1$ and contains no 3-cycle—a contradiction. Hence all counted triples are genuine covers. Each cover has trivial automorphism group: an

automorphism $u \mapsto \kappa u$ acts freely on the m poles and on the $(m+1)/2$ unramified points over c, so its order divides $\gcd(m,(m+1)/2) = 1$.

Now suppose two solutions give isomorphic covers. The isomorphism fixes the unique unramified point over 0 ($u = 0$) and the unique maximally ramified point over c ($u = \infty$). So it is $u \mapsto \kappa u$, and comparing zeros and poles shows the two solutions differ by the torus action. Hence distinct orbits give non-isomorphic covers, and there are at most $m \cdot (\#\text{covers})$ chart points: 3, 10, 35.

For $m = 7$, the exactly verified parametrization (script `verify_msolve_param.py`, shipped with inputs `e5_m7.ms` and `e5_m7.out` in `scripts/`) exhibits 35 distinct points, so the classification is complete in characteristic 0. For $m = 3$ and $m = 5$ the Belyi bound is likewise sharp. In the normalization $\alpha_0 = \beta_0 = \alpha_m = 1$ the system is zero-dimensional, of vector-space dimension 3 for $m = 3$ and 10 for $m = 5$; hence the solution set is nonempty and has at most d points, where $d = 3$ resp. 10, by the dimension alone (independent of the Belyi chart). Eliminating all variables but a_{m-1} gives, up to a nonzero rational factor, the stated eliminant. Each eliminant is irreducible and separable over $\mathbb{Q}$, so the Galois group acts transitively on its d distinct roots; since the system is defined over $\mathbb{Q}$, the solution set is Galois-stable, and every root occurs as the a_{m-1}-coordinate of a solution. Thus there are at least d distinct solutions, and with the upper bound of d, exactly d. All of this is checked by script `e5_exact_counts_m35.py` in `scripts/`. Both classifications are also kernel-checked in Lean (`lean/Jacobian/B26.lean`, over any field of characteristic 0): in this chart the system holds if and only if $T(a_{m-1}) = 0$ and the other unknowns are explicit functions of a_{m-1} (`m3_chart_iff`: $8a_1 = 5a_2^2$; `m5_chart_iff`: $362a_4a_1 = 3a_4^5 + 13078$, $181a_4^2a_2 = 15a_4^5 + 32448$, $2172a_4^3a_3 = 861a_4^5 + 492128$; in both cases β is an explicit polynomial in $a_1, \dots, a_{m-1}$), and T has no repeated root (`m3_T_squarefree`, `m5_T_squarefree`). This gives exactly d solutions over $\overline{\mathbb{Q}}$ without the Galois argument. The count is kernel-checked as well, over every algebraically closed field of characteristic 0 (`m3_chart_card`, `m5_chart_card` in `lean/Jacobian/B26Count.lean`).

The $m = 3$ eliminant $3a_2^3 - 32$ and $m = 5$ eliminant $9a_4^{10} + 37200a_4^5 + 95051008$ are verified irreducible over $\mathbb{Q}$ by factorization (same script). Their irreducibility is also kernel-checked in Lean (`T3poly_irreducible`, `T5poly_irreducible` in `lean/Jacobian/B26Irred.lean`). A rational root of $3a_2^3 - 32$ would satisfy $a_2^3 = 32/3$, but the 3-adic valuation -1 of $32/3$ is not a multiple of 3; a cubic without a rational root is irreducible. For $m = 5$ write the eliminant as $F(a_4^5)$ with $F(u) = 9u^2 + 37200u + 95051008 = (3u + 6200)^2 + 56611008$. This quadratic has no real root, so it is irreducible over $\mathbb{Q}$. Let u be a root of F. The norm of u from $\mathbb{Q}(u)$ to $\mathbb{Q}$ is $95051008/9 = 2^8 \cdot 13^5/3^2$, with 3-adic valuation -2. The norm of a fifth power is a fifth power, and -2 is not a multiple of 5, so u is not a fifth power in $\mathbb{Q}(u)$. Hence $X^5 - u$ is irreducible over $\mathbb{Q}(u)$, by the criterion for $X^p - a$ with p prime. Let $\theta^5 = u$. Then $\mathbb{Q}(\theta)$ contains $u = \theta^5$, so it equals $\mathbb{Q}(u)(\theta)$ and has degree $2 \cdot 5 = 10$ over $\mathbb{Q}$. Since θ is a root of the eliminant, which has degree 10, the eliminant is irreducible. For $m = 7$, the eliminating polynomial in a_6 has degree 35 (Gröbner elimination via SINGULAR `eliminate`). □

Remark 6.2 (Status of `ChartClassification`). `ChartClassification` (in `Jacobian/BranchAbChart.lean`) encodes the 17 equations of the E_5 chart system: the 16 bilinear equations plus the torus normalization $a_7^3 b_0^2 = 1$. Up to v16 of the artifact it was an unproven hypothesis standing for the case $m = 7$ of Proposition 6.1. It is now a theorem, `chartClassification_holds` (§6.4), so the implication `ChartClassification` $L \to \neg\exists \dots$ of `main_theorem_of_chart` is discharged and cannot be vacuous. The statement itself is unchanged. Its 17 equations were verified against the `chart_system_Q.sing` Singular script and by the mechanical chart audit (see the computational archive). The generator of the proof

in §6.4 reads this same statement from the Lean source.

Listing 7: Mechanically extracted from ***Jacobian/BranchAbChart.lean*** *(sha256* ***15907ffeffb6d274****); statement verbatim, proof elided.*

```
theorem main_theorem_of_chart {L : Type*} [Field L] [CharZero L] (hc :
 ↪  ChartClassification L) :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

The two definitions on which `main_theorem_of_chart` rests are printed verbatim in full:

Listing 8: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
abbrev mono (i j : Nat) : Fin 2 →₀ Nat := Finsupp.single 0 i + Finsupp.single 1 j

@[simp] lemma mono_apply_zero (i j : Nat) : mono i j 0 = i := by simp [mono]
@[simp] lemma mono_apply_one (i j : Nat) : mono i j 1 = j := by simp [mono]
```

Listing 9: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def inNP (m : Fin 2 →₀ Nat) : Prop := m 0 ≤ 8 /\ m 1 ≤ 2 * m 0 /\ 2 * m 0 ≤ m 1 + 2
```

Listing 10: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def inNQ (m : Fin 2 →₀ Nat) : Prop := m 0 ≤ 12 /\ m 1 ≤ 2 * m 0 /\ 2 * m 0 ≤ m 1 + 3
 ↪  /\ m 0 ≤ 2 * m 1
```

Listing 11: Mechanically extracted from ***Jacobian/BranchAbNewton.lean*** *(sha256* ***fda6fce4c16abbe8****); definition printed verbatim in full.*

```
def NewtonNF2 (P Q : MvPolynomial (Fin 2) K) : Prop :=
  (forall m ∈ P.support, inNP m) /\ (forall m ∈ Q.support, inNQ m) /\
  P.coeff (mono 0 0) ≠ 0 /\ P.coeff (mono 1 0) ≠ 0 /\ P.coeff (mono 8 14) ≠ 0 /\
   ↪  P.coeff (mono 8 16) ≠ 0 /\
  Q.coeff (mono 0 0) ≠ 0 /\ Q.coeff (mono 2 1) ≠ 0 /\ Q.coeff (mono 12 21) ≠ 0 /\
   ↪  Q.coeff (mono 12 24) ≠ 0
```

Listing 12: Mechanically extracted from ***Jacobian/BranchAbChart.lean*** *(sha256* ***15907ffeffb6d274****); definition printed verbatim in full.*

```
def ChartClassification (L : Type*) [Field L] : Prop :=
  forall a1 a2 a3 a4 a5 a6 a7 b0 b1 b2 b3 b4 b5 b6 b7 b8 b9 : L,
    (3 : L) * b1 + (-2 : L) * a1 * b0 = 0 →
    (5 : L) * b2 + (-5 : L) * a2 * b0 = 0 →
    (7 : L) * b3 + (2 : L) * a1 * b2 + (-3 : L) * a2 * b1 + (-8 : L) * a3 * b0 = 0 →
    (9 : L) * b4 + (4 : L) * a1 * b3 + (-1 : L) * a2 * b2 + (-6 : L) * a3 * b1 + (-11
     ↪  : L) * a4 * b0 = 0 →
    (11 : L) * b5 + (6 : L) * a1 * b4 + (1 : L) * a2 * b3 + (-4 : L) * a3 * b2 + (-9
     ↪  : L) * a4 * b1 + (-14 : L) * a5 * b0 = 0 →
    (13 : L) * b6 + (8 : L) * a1 * b5 + (3 : L) * a2 * b4 + (-2 : L) * a3 * b3 + (-7
     ↪  : L) * a4 * b2 + (-12 : L) * a5 * b1 + (-17 : L) * a6 * b0 = 0 →
    (15 : L) * b7 + (10 : L) * a1 * b6 + (5 : L) * a2 * b5 + (-5 : L) * a4 * b3 +
     ↪  (-10 : L) * a5 * b2 + (-15 : L) * a6 * b1 + (-20 : L) * a7 * b0 = 0 →
    (17 : L) * b8 + (12 : L) * a1 * b7 + (7 : L) * a2 * b6 + (2 : L) * a3 * b5 + (-3
     ↪  : L) * a4 * b4 + (-8 : L) * a5 * b3 + (-13 : L) * a6 * b2 + (-18 : L) * a7 *
     ↪  b1 = 0 →
    (19 : L) * b9 + (14 : L) * a1 * b8 + (9 : L) * a2 * b7 + (4 : L) * a3 * b6 + (-1
     ↪  : L) * a4 * b5 + (-6 : L) * a5 * b4 + (-11 : L) * a6 * b3 + (-16 : L) * a7 *
     ↪  b2 = 0 →
    (21 : L) + (16 : L) * a1 * b9 + (11 : L) * a2 * b8 + (6 : L) * a3 * b7 + (1 : L) *
     ↪  a4 * b6 + (-4 : L) * a5 * b5 + (-9 : L) * a6 * b4 + (-14 : L) * a7 * b3 = 0 →
    (18 : L) * a1 + (13 : L) * a2 * b9 + (8 : L) * a3 * b8 + (3 : L) * a4 * b7 + (-2
     ↪  : L) * a5 * b6 + (-7 : L) * a6 * b5 + (-12 : L) * a7 * b4 = 0 →
    (15 : L) * a2 + (10 : L) * a3 * b9 + (5 : L) * a4 * b8 + (-5 : L) * a6 * b6 +
     ↪  (-10 : L) * a7 * b5 = 0 →
    (12 : L) * a3 + (7 : L) * a4 * b9 + (2 : L) * a5 * b8 + (-3 : L) * a6 * b7 + (-8
     ↪  : L) * a7 * b6 = 0 →
    (9 : L) * a4 + (4 : L) * a5 * b9 + (-1 : L) * a6 * b8 + (-6 : L) * a7 * b7 = 0 →
    (6 : L) * a5 + (1 : L) * a6 * b9 + (-4 : L) * a7 * b8 = 0 →
    (3 : L) * a6 + (-2 : L) * a7 * b9 = 0 →
    a7 ^ 3 * b0 ^ 2 = 1 →
    exists w : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 /\
      a1 = (((512502980) / 1428159681 : L) + ((8080) / 109858437 : L) * w +
       ↪  ((877688) / 1428159681 : L) * w ^ (2 : Nat) + ((127760) / 84009393 : L) *
       ↪  w ^ (3 : Nat) + ((-531068) / 1428159681 : L) * w ^ (4 : Nat)) /\
      a2 = (((7293893720809472) / 226626674937113529 : L) + ((285855137792) /
       ↪  5810940383002911 : L) * w + ((-47390789046272) / 226626674937113529 : L) *
       ↪  w ^ (2 : Nat) + ((-90487897751552) / 226626674937113529 : L) * w ^ (3 :
       ↪  Nat) + ((21860765739008) / 226626674937113529 : L) * w ^ (4 : Nat)) /\
      a3 = (((6464293233975242927636 48) / 32365907978427875263742 4249 : L) +
       ↪  ((-311599571772213100544) / 24896852291098365587494173 : L) * w +
       ↪  ((-13107803800410363756544) / 32365907978427875263742 4249 : L) * w ^ (2 :
       ↪  Nat) + ((-39360612513016248860672) / 32365907978427875263742 4249 : L) * w
       ↪  ^ (3 : Nat) + ((988619118934257079 0912) / 32365907978427875263742 4249 : L)
       ↪  * w ^ (4 : Nat)) /\
      a4 = (((401489080334849991328866 1860352) /
       ↪  154078949379156364060580574704501523 : L) +
       ↪  ((-19247583898558620695724556288) / 39507422917732401041174506334 48757 :
       ↪  L) * w + ((-14781345296916798503 1874871296) /
       ↪  154078949379156364060580574704501523 : L) * w ^ (2 : Nat) +
       ↪  ((-21329313178098532487186612 22400) /
       ↪  154078949379156364060580574704501523 : L) * w ^ (3 : Nat) +
       ↪  ((58039857816239632626286055178240) / 154078949379156364060580574704501523
       ↪  : L) * w ^ (4 : Nat)) /\
      a5 = (((-47195336258536808324667608668861104128) /
       ↪  7334978106471703364862620608159252145056 4721 : L) +
       ↪  ((-29932787320250950904651205753668894 72) /
       ↪  5642290851132079511432785083199424726966517 : L) * w +
       ↪  ((32500098714604785191372567805805199 36) /
       ↪  8149975673857448183180689564621391272284969 : L) * w ^ (2 : Nat) +
       ↪  ((-35972900451489262851635484544208994304) /
       ↪  7334978106471703364862620608159252145056 4721 : L) * w ^ (3 : Nat) +
       ↪  ((42124958173030895116478408190549032 96) /
       ↪  2444992702157234454954206869386417381685 4907 : L) * w ^ (4 : Nat)) /\
```

```
a6 = (((-353637771581420436331059704199622022880296 96) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) +
 ↪  ((-244358877861941369482776189370917360542679 04) /
 ↪  185955976201430980705718819939161127941107992686 1787 : L) * w +
 ↪  ((348725961520751776828947066063346243175710720) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) * w ^ (2 : Nat)
 ↪  + ((-891434267787294593043481518636230483797606 4) /
 ↪  142201628859917808774961450541711450778494347348 2543 : L) * w ^ (3 : Nat)
 ↪  + ((677989855609902867411530105006060279347281 92) /
 ↪  241742769061860274917434465920909466323440390492032 31 : L) * w ^ (4 :
 ↪  Nat)) /\
a7 = (((342097029127255765817099097920847945090722114306048) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) +
 ↪  ((-568105419722006463099987956174489022887517749248 0) /
 ↪  226987032181093385596749183555563073408523912496789 14982991 : L) * w +
 ↪  ((206885581446139393490715159158183040429422553333760) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (2
 ↪  : Nat) + ((-633482589598705098139540581219264850266194790 31808) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (3
 ↪  : Nat) + ((253322561059155215789084089746168458666056079 11424) /
 ↪  885249425506264203827321815866695986293243258737477684336649 : L) * w ^ (4
 ↪  : Nat)) /\
b0 = (((241787651899027059649022111227674440235744649167926923464101195333242
    8638485407) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) +
 ↪  ((147114702866918150301106182425486691319761000221649016655576552446507 3
 ↪  2041478549) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w + ((617854596280970291332161410119563350111260243577025705635188990137 8
 ↪  76816776295) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w ^ (2 : Nat) + ((-366378570719933994254445864210576333868526055032758879
 ↪  4221990329866702931989049) /
 ↪  811441542599883814485270953564641654639931576344287204467623854 08 : L) *
 ↪  w ^ (3 : Nat) + ((283919616043133221256996475572763279716168016664606915 3
 ↪  9514166693155875345 3503) /
 ↪  405720771299941907242635476782320827319965788172143602233811927 04 : L) *
 ↪  w ^ (4 : Nat)) /\
b1 = (((213129345216263992177813311926195028654591343373099779726908014927974
    58781) / 206452661968217945879623181753674347303056069698831468671795 2 :
 ↪  L) + ((907860591902046261556432976412843546973938642324006831011933983788
 ↪  54556479) / 206452661968217945879623181753674347303056069698831468671795 2
 ↪  : L) * w + ((367683922090893100561414709129515035875503403273788620768039
 ↪  4068928684725) /
 ↪  206452661968217945879623181753674347303056069698831468671795 2 : L) * w ^
 ↪  (2 : Nat) + ((-210875061122372142827023757868450079885049551692627677710 7
 ↪  9750788915150635) /
 ↪  206452661968217945879623181753674347303056069698831468671795 2 : L) * w ^
 ↪  (3 : Nat) + ((180447192994234227283436464239178979183551309852872229435 04
 ↪  00675635340349) /
 ↪  103226330984108972939811590876837173651528034849415734335897 6 : L) * w ^
 ↪  (4 : Nat)) /\
```

```
b2 = (((-144544033917986925185258501166756369390602635518529573940467345287) /
 ↪  4103682631861140601935060318162479234442106514660392960 : L) +
 ↪  ((575311520507585192568556318243232964279467466281194477346330819301) /
 ↪  1025920657965285150483765079540619808610526628665098240 : L) * w +
 ↪  ((5433069126889736877942221681516595789619526476032859426349490049 5) /
 ↪  2051841315930570300967530159081239617221053257330196480 : L) * w ^ (2 :
 ↪  Nat) +
 ↪  ((-1741878800183793363061982639010592400471804726492262030581281408 95) /
 ↪  1025920657965285150483765079540619808610526628665098240 : L) * w ^ (3 :
 ↪  Nat) +
 ↪  ((8016655714033798990913285654216147452190928620315580375862611003 3) /
 ↪  4103682631861140601935060318162479234442106514660392960 : L) * w ^ (4 :
 ↪  Nat)) /\
b3 = (((-7425458405034303101010032795601866242474210402999635943131) /
 ↪  2039233955921481334763488368591261005182612275 2 : L) +
 ↪  ((6744315542914009035390981237841878170110250243622814128087) /
 ↪  2039233955921481334763488368591261005182612275 2 : L) * w +
 ↪  ((4784407492357442361348728810064362618273934934160988178 53) /
 ↪  2039233955921481334763488368591261005182612275 2 : L) * w ^ (2 : Nat) +
 ↪  ((-3700344121490310017718636169454092008434178658425433227171) /
 ↪  2039233955921481334763488368591261005182612275 2 : L) * w ^ (3 : Nat) +
 ↪  ((6223309793893257222270260849950285980688513443063940211 7) /
 ↪  1019616977960740667381744184295630502591306137 6 : L) * w ^ (4 : Nat)) /\
b4 = (((-1969821243345379668348779806541297557387675803172 47) /
 ↪  4053408121472769419865599653882359709696 : L) +
 ↪  ((7356162939037843464175425549716484499362290432253) /
 ↪  1013352030368192354966399913470589927424 : L) * w +
 ↪  ((3334214783216888283093389346861252319817155285863) /
 ↪  2026704060736384709932799826941179854848 : L) * w ^ (2 : Nat) +
 ↪  ((-1359277153685966616685055654343736877744492161395 9) /
 ↪  1013352030368192354966399913470589927424 : L) * w ^ (3 : Nat) +
 ↪  ((-1743167522426431101053174423220977563711561703271) /
 ↪  4053408121472769419865599653882359709696 : L) * w ^ (4 : Nat)) /\
b5 = (((-5945326279911459300427967186228236954680873) /
 ↪  1611400923519540560385711240380416 : L) +
 ↪  ((-1207414796044598637778246641365546517002499) /
 ↪  1611400923519540560385711240380416 : L) * w +
 ↪  ((1473979068322069070539748868284852592790 3) /
 ↪  1611400923519540560385711240380416 : L) * w ^ (2 : Nat) +
 ↪  ((-1003046473955460175839195938091594613306305) /
 ↪  1611400923519540560385711240380416 : L) * w ^ (3 : Nat) +
 ↪  ((-5798945797632118042689943053535315371961481) /
 ↪  8057004617597702801928556201902 08 : L) * w ^ (4 : Nat)) /\
b6 = (((-4071756407934756085572029277992757 3) / 3202999622174946730107207 68 :
 ↪  L) + ((-4431956341187962499826675611271825) / 8007499055437366825268019 2 :
 ↪  L) * w + ((7557473379676168234493663207109 57) /
 ↪  1601499811087473365053603 84 : L) * w ^ (2 : Nat) +
 ↪  ((-8694712748920386764741075145911 97) / 8007499055437366825268019 2 : L) *
 ↪  w ^ (3 : Nat) + ((-9388385777777770165473241283459 97) /
 ↪  3202999622174946730107207 68 : L) * w ^ (4 : Nat)) /\
b7 = (((-4225899495200050988086279 5) / 31833190703543877632 : L) +
 ↪  ((-3637618801445519626 5319 4745) / 31833190703543877632 : L) * w +
 ↪  ((5527319619223464764982093) / 31833190703543877632 : L) * w ^ (2 : Nat) +
 ↪  ((3622463246490253699729005) / 31833190703543877632 : L) * w ^ (3 : Nat) +
 ↪  ((-7170396348706113287216 43) / 15916595351771938816 : L) * w ^ (4 : Nat))
 ↪  /\
b8 = (((89529406222349457) / 6327518887936 : L) + ((-9772922162070171) /
 ↪  1581879721984 : L) * w + ((8308630462897287) / 3163759443968 : L) * w ^ (2
 ↪  : Nat) + ((8039403284956497) / 1581879721984 : L) * w ^ (3 : Nat) +
 ↪  ((-2811959945010567) / 6327518887936 : L) * w ^ (4 : Nat)) /\
b9 = (((1461767229) / 5030912 : L) + ((82735263) / 5030912 : L) * w +
 ↪  ((34254981) / 5030912 : L) * w ^ (2 : Nat) + ((182339829) / 5030912 : L) *
 ↪  w ^ (3 : Nat) + ((-10377315) / 2515456 : L) * w ^ (4 : Nat))
```

The top-layer classification statement and its witnesses:

*Listing 13: Mechanically extracted from **Jacobian/BranchAbClassification.lean** (sha256 **29fdecd5a9fda336**); definition printed verbatim in full.*

```
def TopLayerClassification (L : Type*) [Field L] : Prop :=
  forall (A₂ B₃ : L[X]) (lam : L),
    (forall i, A₂.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8) → (forall i, B₃.coeff i ≠ 0 → 2 ≤ i
     ↪  /\ i ≤ 12) →
    A₂.coeff 1 ≠ 0 → A₂.coeff 8 ≠ 0 → B₃.coeff 2 ≠ 0 → B₃.coeff 12 ≠ 0 →
    layerTerm 2 3 A₂ B₃ = C lam * X ^ 2 →
    exists w ρ σ ε : L, w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0 /\ ρ ≠ 0 /\ σ ≠
     ↪  0 /\ ε ≠ 0 /\
      (forall i, A₂.coeff i = ρ * ε ^ (i - 1) * topA w i) /\ (forall k, B₃.coeff k =
       ↪  σ * ε ^ (k - 2) * topB w k)
```

*Listing 14: Mechanically extracted from **Jacobian/BranchAbSharp.lean** (sha256 **9eb4eb229d2047f6**); definition printed verbatim in full.*

```
noncomputable def topA {L : Type*} [Field L] (w : L) : Nat → L
  | 1 => ((1 : L))
  | 2 => (((303) / 16 : L) + ((879) / 128 : L) * w + ((-389) / 128 : L) * w ^ (2 :
   ↪  Nat) + ((-51) / 128 : L) * w ^ (3 : Nat) + ((49) / 128 : L) * w ^ (4 : Nat))
  | 3 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : Nat) +
   ↪  ((-189) / 8 : L) * w ^ (3 : Nat) + ((147) / 16 : L) * w ^ (4 : Nat))
  | 4 => (((-79653) / 64 : L) + ((29463) / 64 : L) * w + ((7371) / 128 : L) * w ^ (2 :
   ↪  Nat) + ((-4767) / 16 : L) * w ^ (3 : Nat) + ((8001) / 128 : L) * w ^ (4 : Nat))
  | 5 => (((-33519) / 4 : L) + ((-6843) / 4 : L) * w + ((8181) / 8 : L) * w ^ (2 :
   ↪  Nat) + ((-1713) / 2 : L) * w ^ (3 : Nat) + ((-1305) / 8 : L) * w ^ (4 : Nat))
  | 6 => (((-8805) / 8 : L) + ((-76965) / 8 : L) * w + ((11523) / 16 : L) * w ^ (2 :
   ↪  Nat) + ((6927) / 4 : L) * w ^ (3 : Nat) + ((-18579) / 16 : L) * w ^ (4 : Nat))
  | 7 => (((389740) / 17 : L) + ((-51922) / 17 : L) * w + ((-48572) / 17 : L) * w ^
   ↪  (2 : Nat) + ((63239) / 17 : L) * w ^ (3 : Nat) + ((-6475) / 17 : L) * w ^ (4 :
   ↪  Nat))
  | 8 => (((476437) / 68 : L) + ((314405) / 68 : L) * w + ((-174571) / 136 : L) * w ^
   ↪  (2 : Nat) + ((1574) / 17 : L) * w ^ (3 : Nat) + ((82783) / 136 : L) * w ^ (4 :
   ↪  Nat))
  | _ => 0
```

*Listing 15: Mechanically extracted from **Jacobian/BranchAbSharp.lean** (sha256 **9eb4eb229d2047f6**); definition printed verbatim in full.*

```
noncomputable def topB {L : Type*} [Field L] (w : L) : Nat → L
  | 2 => ((1 : L))
  | 3 => (((101) / 8 : L) + ((293) / 64 : L) * w + ((-389) / 192 : L) * w ^ (2 : Nat)
   ↪  + ((-17) / 64 : L) * w ^ (3 : Nat) + ((49) / 192 : L) * w ^ (4 : Nat))
  | 4 => (((315) / 16 : L) + ((693) / 8 : L) * w + ((-147) / 8 : L) * w ^ (2 : Nat) +
   ↪  ((-189) / 8 : L) * w ^ (3 : Nat) + ((147) / 16 : L) * w ^ (4 : Nat))
  | 5 => (((-11379) / 8 : L) + ((4209) / 8 : L) * w + ((1053) / 16 : L) * w ^ (2 :
   ↪  Nat) + ((-681) / 2 : L) * w ^ (3 : Nat) + ((1143) / 16 : L) * w ^ (4 : Nat))
  | 6 => (((-695299) / 48 : L) + ((-48367) / 48 : L) * w + ((52679) / 32 : L) * w ^
   ↪  (2 : Nat) + ((-45661) / 24 : L) * w ^ (3 : Nat) + ((-2211) / 32 : L) * w ^ (4 :
   ↪  Nat))
  | 7 => (((-309895) / 8 : L) + ((-251275) / 8 : L) * w + ((101481) / 16 : L) * w ^ (2
   ↪  : Nat) + ((1811) / 4 : L) * w ^ (3 : Nat) + ((-60489) / 16 : L) * w ^ (4 : Nat))
```

```
| 8 => (((1733348) / 13 : L) + ((-2368725) / 26 : L) * w + ((-184976) / 13 : L) * w
↪  ^ (2 : Nat) + ((1819939) / 52 : L) * w ^ (3 : Nat) + ((-569071) / 52 : L) * w ^
↪  (4 : Nat))
| 9 => (((18736247) / 39 : L) + ((1087133) / 13 : L) * w + ((-5656777) / 78 : L) *
↪  w ^ (2 : Nat) + ((1993384) / 39 : L) * w ^ (3 : Nat) + ((971365) / 78 : L) * w
↪  ^ (4 : Nat))
| 10 => (((305098) / 17 : L) + ((5784066) / 17 : L) * w + ((-308335) / 17 : L) * w
↪  ^ (2 : Nat) + ((-950608) / 17 : L) * w ^ (3 : Nat) + ((754021) / 17 : L) * w ^
↪  (4 : Nat))
| 11 => (((-149721182) / 323 : L) + ((26391690) / 323 : L) * w + ((21560033) / 323
↪  : L) * w ^ (2 : Nat) + ((-24378412) / 323 : L) * w ^ (3 : Nat) + ((2786935) /
↪  323 : L) * w ^ (4 : Nat))
| 12 => (((-91912444) / 969 : L) + ((-20715500) / 323 : L) * w + ((17474146) / 969
↪  : L) * w ^ (2 : Nat) + ((-1623428) / 969 : L) * w ^ (3 : Nat) + ((-8783302) /
↪  969 : L) * w ^ (4 : Nat))
| _ => 0
```

6.3 Irreducibility and reality

Theorem 6.3 (Top-layer structure). *Let $W \in \mathbb{Z}[a_6]$ be the degree-35 eliminating polynomial for a_6 at $m = 7$. Then:*

(i) *W is irreducible over $\mathbb{Q}$ (Singular **`factorize`** over $\mathbb{Q}$ yields a nonzero constant times a single degree-35 factor);*
(ii) *$W(a_6) = V(a_6^7)$ for a quintic $V \in \mathbb{Z}[S]$;*
(iii) *V is separable (nonzero discriminant) and has exactly one real root, $S \approx 3208.612478738048$;*
(iv) *W is separable modulo 101 ($\gcd(W, W') = 1$ in $\mathbb{F}_{101}[a_6]$), so the 35 complex solutions are distinct.*

Consequently the 35 solutions form a single Galois orbit over $\mathbb{Q}$, comprising one real solution and 17 complex-conjugate pairs. See Figure 3. The map $a_6 \mapsto a_6^7$ sends the 35 roots of W onto the 5 roots of V, 7 to 1, and both sets are single Galois orbits.

Proof. (i) Direct factorization in Singular over $\mathbb{Q}$. (ii) By inspection: every exponent of W is divisible by 7. (iii) Discriminant computed exactly in SymPy (nonzero integer); real-root count via Sturm sequence gives exactly one real S. (iv) Direct gcd computation in $\mathbb{F}_{101}[a_6]$. □

Remark 6.4 (Characteristic-zero/modular consistency). Reducing the integer quintic V modulo 101 gives 58 times the quintic obtained directly from the mod-101 Gröbner computation (equivalently, the mod-101 quintic is 54 times V), confirming consistency between the characteristic-zero and modular routes.

6.4 An algebraic proof of the top-layer classification for $m = 7$, and its formalization

The proof of Proposition 6.1 above rests on Riemann's existence theorem, which is not in Mathlib (at least not in the version pinned by the artifact). For $m = 7$ we now give a second proof of the classification it provides, in the normalized chart of §5: every solution is one of the points $(P_j(w), Q_k(w))$ with $R(w) = 0$, the five K_5-conjugates. This is the form in which the main theorem uses Proposition 6.1. The eliminant W and the cases $m = 3, 5$ are not reproved here. The proof is purely algebraic and valid over every field of characteristic 0, and it is formalized: Theorem 6.5 is the Lean theorem `chartClassification_holds` (listing 16). We

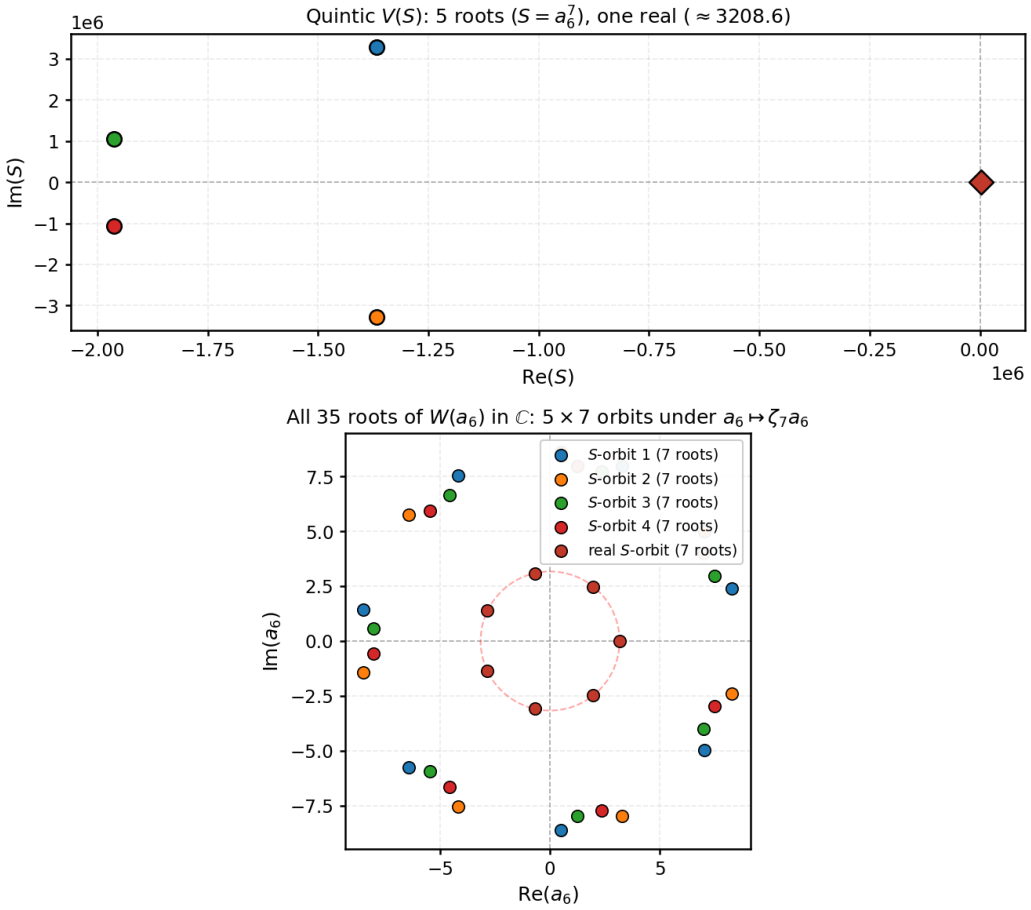


Figure 3: Top: the five roots of the quintic $V(S)$ in the complex S-plane, computed from the integer coefficients; the real root $S \approx 3208.61$ is marked with a diamond. Bottom: all 35 roots of $W(a_6)$ in $\mathbb{C}$, colored by S-orbit; each $S \neq 0$ lifts to 7 values of a_6 via 7th roots of unity. The dashed circle marks the real S-orbit. The 35 roots form a single Galois orbit over $\mathbb{Q}$.

work in the Lean chart of §5: $\alpha = 1 + a_1 u + \cdots + a_7 u^7$, $\beta = b_0 + b_1 u + \cdots + b_9 u^9 + u^{10}$, and $E_n = \sum_{i+k=n} (1 + 2k - 3i)\, a_i b_k$ $(1 \le n \le 16)$ is the coefficient of u^n in $\alpha\beta + u(2\alpha\beta' - 3\alpha'\beta)$.

Theorem 6.5 (`chartClassification_holds`). *Let L be a field of characteristic 0. Suppose $(a_1, \dots, a_7, b_0, \dots, b_9) \in L^{17}$ satisfies $E_1 = \cdots = E_{16} = 0$ and $a_7^3 b_0^2 = 1$. Then $(a_j, b_k) = (P_j(w), Q_k(w))$ for a root $w \in L$ of R. Here $P_j, Q_k \in \mathbb{Q}[w]$ are the coordinates in listing 12.*

Proof. Step 1: reversed coordinates. Since $a_7^3 b_0^2 = 1$, we have $a_7 \neq 0$. Put $y_i = a_{7-i}/a_7$ for $1 \le i \le 6$ and $y_7 = 1/a_7$, and give y_i the weight i. Define $S_k \in \mathbb{Q}[y_1, \dots, y_7]$, weighted homogeneous of weight k, by

$$(1 + y_1 v + \cdots + y_7 v^7)^{3/2} = \sum_{k \ge 0} S_k v^k,$$

or, equivalently, by $S_0 = 1$ and $2k\, S_k = \sum_{i=1}^{7} (5i - 2k)\, y_i\, S_{k-i}$. Write $\tilde\beta_k = b_{10-k}$ for $0 \le k \le 10$ (so $\tilde\beta_0 = 1$) and $\tilde\beta_k = 0$ for $k > 10$; in both recursions, terms with a negative index are zero. Reindexing E_{17-k} by $i \mapsto 7 - i$, $k \mapsto 10 - k$ gives

$$E_{17-k}/a_7 = -2k\,\tilde\beta_k + \sum_{i=1}^{7} (5i - 2k)\, y_i\, \tilde\beta_{k-i} \qquad (1 \le k \le 16).$$

This is the recursion for S_k. Induction on k gives $b_{10-k} = S_k(y)$ for $1 \le k \le 10$, and

$$S_{11}(y) = \cdots = S_{16}(y) = 0, \qquad S_{10}(y)^2 = y_7^3. \tag{7}$$

The last equation holds because $b_0 = S_{10}(y)$ and $a_7^3 b_0^2 = 1$. In the language of §6, the reversed β is the truncation of the power-series square root of the reversed α^3/a_7^3. This is the Davenport–Stothers condition.

Step 2: the orbit relations. Consider the six weighted-homogeneous relations

$$\begin{aligned}
y_4 &= \tfrac{2431}{18240} y_1^4 - \tfrac{1027}{1710} y_1^2 y_2 + \tfrac{143}{380} y_2^2 + \tfrac{41}{60} y_1 y_3, \\
y_5 &= \tfrac{1154351}{25313472} y_1^5 - \tfrac{3833107}{18985104} y_1^3 y_2 + \tfrac{6512}{43947} y_1 y_2^2 + \tfrac{24449}{166536} y_1^2 y_3, \\
y_6 &= \tfrac{4206749569}{4242031637760} y_1^6 + \tfrac{8589189473}{1590761864160} y_1^4 y_2 - \tfrac{440013827}{9819517680} y_1^2 y_2^2 \\
&\quad + \tfrac{7454351}{734423760} y_1^3 y_3 + \tfrac{253}{5054} y_2^3, \\
y_7 &= -\tfrac{4539711243127}{16353031963564800} y_1^7 + \tfrac{13373800868041}{6132386986336800} y_1^5 y_2 - \tfrac{214713769679}{3785424065640 0} y_1^3 y_2^2 \\
&\quad + \tfrac{81661753}{707800898700} y_1^4 y_3 + \tfrac{92333}{19483170} y_1 y_2^3, \\
y_2 y_3 &= \tfrac{3738317}{46630080} y_1^5 - \tfrac{16882087}{34972560} y_1^3 y_2 + \tfrac{115021}{161910} y_1 y_2^2 + \tfrac{2277881}{5828760} y_1^2 y_3, \\
y_3^2 &= \tfrac{740761013029}{10605079094400} y_1^6 - \tfrac{754440973541}{1988452330200} y_1^4 y_2 + \tfrac{3827878261}{8182931400} y_1^2 y_2^2 \\
&\quad + \tfrac{137816863}{918029700} y_1^3 y_3 + \tfrac{289}{12635} y_2^3.
\end{aligned}$$

Write $g = 0$ for any of them. Each satisfies

$$y_7^3 \cdot g \in (S_{11}, \dots, S_{16}) \subset \mathbb{Q}[y_1, \dots, y_7],$$

with explicit cofactors of 341 to 694 terms, whose numerators and denominators have at most 42 digits. Since $y_7 \neq 0$, every solution of (7) satisfies all six relations.

The relations were found by exact interpolation on the K_5 point, as all weighted-homogeneous forms of weight at most 7 that vanish on the five orbits. The cofactors come from exact weighted-homogeneous linear algebra over $\mathbb{Q}$. Let (g) be the ideal of the six relations. A Gröbner computation over $\mathbb{Q}$ (`saturation_check.py`) shows that $(S_{11},\dots,S_{16}) \subseteq (g)$ and $(g) : y_7 = (g)$. With the certificates this gives

$$(S_{11},\dots,S_{16}) : y_7^{\infty} = (S_{11},\dots,S_{16}) : y_7^{3} = (g).$$

The ideal (g) has Krull dimension 1 and needs all six generators, so it is a weighted complete intersection. It has exactly five solutions in the chart $y_1 = 1$ and none with $y_1 = 0$ other than $y = 0$, in agreement with weighted Bézout, $4 \cdot 5 \cdot 5 \cdot 6 \cdot 6 \cdot 7/7! = 5$. These facts explain the choice of relations but are not used in the proof.

Step 3: a resultant. The relation for y_7 shows $y_7 \in y_1\,\mathbb{Q}[y_1, y_2, y_3]$, so $y_1 \neq 0$. Put $t = y_2/y_1^2$ and $s = y_3/y_1^3$. The relation for y_2y_3 becomes $(t - c)\,s = N(t)$ with $c = \frac{2277881}{5828760}$ and $\deg N = 2$, and the relation for y_3^2 becomes monic quadratic in s. Their Sylvester resultant in s is $\kappa\, m(t)$ with $\kappa \in \mathbb{Q}^{\times}$ and

$$\begin{aligned} m(t) = {} & 287548593020928\,t^5 - 688401965085696\,t^4 + 640652914818432\,t^3 \\ & - 292066554895024\,t^2 + 65563255857792\,t - 5817852446211, \end{aligned}$$

which is irreducible over $\mathbb{Q}$. Hence $t - c$ is invertible modulo m, and $s = q_3(t)$. The first four relations then give $y_i/y_1^i = q_i(t)$ for $4 \le i \le 7$, with $q_i \in \mathbb{Q}[t]$ reduced modulo m.

Step 4: the torus. Let $z = 1/y_1$. By weighted homogeneity,

$$z^{10} S_{10}(y) = S_{10}\bigl(1, t, q_3(t), \dots, q_7(t)\bigr) \equiv \sigma(t) \pmod{m} \qquad \text{and} \qquad z^7 y_7 = q_7(t).$$

Multiplying $S_{10}^2 = y_7^3$ by z^{21} gives $z\,\sigma(t)^2 = q_7(t)^3$, that is,

$$y_1 = \sigma(t)^2\, q_7(t)^{-3} \bmod m(t).$$

This fixes the torus without a seventh root, as in §5, and makes every y_i a polynomial in t modulo m. The substitution $w = \omega(t)$, with $\omega \in \mathbb{Q}[t]$ of degree 4, induces an isomorphism $\mathbb{Q}[t]/(m) \cong \mathbb{Q}[w]/(R) = K_5$ and gives $R(w) = 0$ and $y_i = Y_i(w)$.

Step 5: back to the chart. The formulas $a_7 = 1/y_7$ and $a_{7-i} = y_i/y_7$, together with the recursion of Step 1 for b_{10-k}, give $a_j = P_j(w)$ and $b_k = Q_k(w)$. □

Lemma 6.6 (Characteristic-zero transfer)**.** *Let F be a field of characteristic 0. Every F-valued solution of the normalized $m = 7$ chart system is obtained by specializing the K_5-point at a root $w \in F$ of R; its five specializations at the five roots of R (over a splitting field of R) are the five solutions.*

Proof. Steps 1–5 are proved algebraically over $\mathbb{Q}$ and hence hold over every characteristic-zero field by base change. The Lean formalization quantifies over all L with `CharZero`. The five solutions are the K_5-conjugates $(P_j(w), Q_k(w))$ with $R(w) = 0$. □

Remark 6.7 (Roles of the two classifications)*.* The $m = 7$ case of Proposition 6.1 is proved twice. The Belyi route combines the geometric upper bound (at most 35 chart points over $\mathbb{C}$) with the 35 exhibited points from the msolve parametrization, giving a complete classification in characteristic 0; it depends on Riemann's existence theorem and the computed parametrization. The algebraic certificate of this section gives an independent classification: every solution in characteristic 0 is one of the five K_5-orbits (Lemma 6.6). Lean's `main_theorem` uses only the algebraic route.

Remark 6.8 (Formalization). The proof is formalized in `lean/Jacobian/ChartProof/`: **12** modules, all generated by `lean/certgen/chartproof/` except the hand-written `Reflect.lean`.

- *Steps.* There are **111** steps. In **105** of them a polynomial g vanishes because $g = \sum_i c_i e_i$ for polynomials e_i already known to vanish. In the other six, one for each relation of Step 2, the identity $y_7^3 g = e$ holds with e already known to vanish, and y_7^3 is cancelled using $y_7 \neq 0$.
- *Kernel checking.* Every identity is checked by the Lean kernel. `lc_zero` (listing 18) and `eq_of_toPolyK` reduce it to an equality of normal forms. These are computed by the verified normalizer of `Lean.Grind.CommRing` from Lean core, with a kernel-friendly multiplication defined by recursors (`toPolyK`, soundness theorem `denote_toPolyK`), and `decide +kernel` evaluates them.
- *Axioms.* No `native_decide` is used. `#print axioms` for `chartClassification_holds` and for `main_theorem` reports only `propext`, `Classical.choice` and `Quot.sound`.
- *Batching.* The certificates of Step 2 have between **27,747** and **56,999** monomial products each. They are split into batches of at most **9,000** products, one declaration per batch.
- *Independent checks.* Every certificate was also verified by exact SymPy expansion before it was written, and the generator reproduces all eleven generated modules byte-identically. A second checker, written independently of the generator (`independent_check.py`), parses the Lean text and re-verifies all **111** identities over $\mathbb{Z}$. This checker verifies the arithmetic of the certificates only; it does not re-prove the mathematical statements. The Lean kernel remains solely responsible for verifying that these certificates prove the stated formal propositions. Six controls behave as expected: four one-integer perturbations are rejected by `decide`, a planted `sorry` is flagged, and an unmodified module is accepted.
- *Cost.* The build takes about **25** minutes sequentially and needs at most **5.4** GB of memory for any one module.

Remark 6.9 (Why these coordinates). Earlier formulations failed to yield certificates of practical size because of the degenerate Davenport–Stothers components of lower degree (the solutions for $m = 1, 3, 5$). With $a_0 = b_0 = 1$ and b eliminated, these components force saturation by a_7^7 or a_7^8, so the certificates have weight **61**. In the chart $a_0 = b_{10} = 1$, the sub-case $a_1 = 0$ alone needed **503,254** cofactor terms modulo a prime. Replayed over $\mathbb{Q}$, its Buchberger trace reached coefficients of **2,722** digits after **929** of its **2,065** elements (`lean/CLASSIFICATION_STATUS.md`, §5). In the reversed coordinates the same components lie on $y_7 = 0$, and y_7^3 suffices, at weight at most **28**. The five orbits then form a weighted complete intersection that can be written down explicitly. The six certificates of Step 2 have **341** to **694** cofactor terms each over $\mathbb{Q}$, roughly a thousand times fewer than the modular lift for $a_1 = 0$ alone.

Listing 16: Mechanically extracted from ***Jacobian/ChartProof/Final.lean*** *(sha256* ***638b644ffb58cb8b****); statement verbatim, proof elided.*

```
theorem chartClassification_holds (L : Type*) [Field L] [CharZero L] :
↪  ChartClassification L := by
  -- proof body elided (see artifact)
```

Listing 17: Mechanically extracted from ***Jacobian/ChartProof/Final.lean*** *(sha256* ***638b644ffb58cb8b****); statement verbatim, proof elided.*

```
theorem main_theorem {L : Type*} [Field L] [CharZero L] :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 :=
  -- proof term elided (see artifact)
```

Listing 18: Mechanically extracted from ***Jacobian/ChartProof/Reflect.lean*** *(sha256* ***9e3ea428ffe7236b****); statement verbatim, proof elided.*

```
theorem lc_zero (ctx : Context α) (g : Expr) (l : List (Expr × Expr))
    (hcert : Poly.beq' (toPolyK (.sub g (lcExpr l))) (.num 0) = true) (h : AllZero
     ↪  ctx l) :
    g.denote ctx = 0 := by
  -- proof body elided (see artifact)
```

7 Descent

Fix a top-layer solution (α, β). The descent solves E_4, E_3, E_2 in turn. We use the scaled variables $A_2 = u\alpha$ and $B_3 = u^2\beta$.

7.1 Layer E_4

Proposition 7.1 (E_4 structure)**.** *E_4 is a linear system in 19 unknowns (A_1 on $u^1, \dots, u^8$; B_2 on $u^2, \dots, u^{12}$) with 18 equations, rank 17, and kernel dimension 2. The kernel is parametrized by $t = (t_1, t_2)$, giving $(A_1(t), B_2(t))$ linear in t. Computed exactly in characteristic zero over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script* ***exact_ranks_K5.py****); the modular evaluation at $p = 101$, $a_6 = 55$ is an independent cross-check.*

Proof. Rank computation on the 18×19 coefficient matrix, with residual asserts verifying the kernel. Scripts: `exact_ranks_K5.py` (exact rank and kernel over K_5) and `C5_E2_obstruction_corrected.py` (which verifies E_4 with E_4-kernel residuals zero; mod-101 cross-check). □

7.2 Layer E_3

Proposition 7.2 (E_3 structure and unconditional solvability)**.** *E_3 is a 19×20 linear system of rank 18 with 2-dimensional kernel. Moreover, E_3 is solvable for* every *direction $t = (t_1, t_2)$: writing the system as $M(t)x = K(t)$ with M constant and K quadratic in t, the left nullvector v (with $v^{\mathsf{T}}M = 0$) satisfies*

$$v^{\mathsf{T}}K_{11} \equiv 0, \qquad v^{\mathsf{T}}K_{22} \equiv 0, \qquad v^{\mathsf{T}}K_{12} \equiv 0 \tag{8}$$

identically as polynomials in t, where $K = t_1^2K_{11} + t_1t_2K_{12} + t_2^2K_{22}$. Hence no direction is obstructed at E_3. All computations are exact over K_5; see Lemma 9.2(i).

Proof. The 19×20 matrix M has rank 18; its left nullspace is 1-dimensional, spanned by v. Decomposing the inhomogeneous part and computing $v^{\mathsf{T}} K_{ij}$ symbolically gives zero for all i, j. Script: `C5_E2_obstruction_corrected.py` (which verifies E_3 with residual asserts). □

The general solution of E_3 is thus the 2-parameter affine family

$$x(t,s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2, \tag{9}$$

encoding (A_0, B_1) in terms of $(t,s) = (t_1, t_2, s_1, s_2) \in \mathbb{C}^4$.

7.3 Layer E_2

Proposition 7.3 (E_2 injectivity)**.** *The homogeneous operator of E_2 in the unknown B_0' is a 19×12 matrix of rank 12, hence injective. Its left nullspace has dimension $19 - 12 = 7$. Exact over K_5; see Lemma 9.2(i)–(ii).*

Therefore B_0' (and hence B_0 up to an additive constant) is uniquely fixed by (A_0, B_1, A_1, B_2). The seven left-nullvector pairings then give seven compatibility conditions:

$$W \cdot \mathrm{rhs}(t,s) = 0, \tag{10}$$

where W is 7×19 and $\mathrm{rhs}(t,s)$ is the E_2 inhomogeneous term. These seven conditions are the *only* remaining obstructions; E_1 is never needed.

Listing 19: Mechanically extracted from ***`Jacobian/Descent/Main.lean`*** *(sha256* ***`e18a3a9e531df6d6`****); statement verbatim, proof elided.*

```
theorem descent_K5 {L : Type*} [Field L] [CharZero L]
    (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
    (a_1_0 a_2_2 a_3_4 a_4_6 a_5_8 a_6_10 a_7_12 a_8_14 b_2_1 b_3_3 b_4_5 b_5_7
     ↪  b_6_9 b_7_11 b_8_13 b_9_15 b_10_17 b_11_19 b_12_21 : L)
    (h_a_1_0 : a_1_0 = ((1 : L)))
    -- ... (18 h4_*, 19 h3_*, 19 h2_*, 8 h_a_*, 11 h_b_*, 51 lower-layer unknowns in
     ↪  one binder)
    (hv : b_12_24 ≠ 0) :
    False := by
  -- proof body elided (see artifact)
```

8 The final obstruction

We use the homogeneous formulation: E_3 with A_0 on $u^1, \ldots, u^8$ ($a_{8,16}$ free, no constant term) and B_1 on $u^1, \ldots, u^{12}$. This gives the genuine 2-dimensional kernel with two independent directions $s = (s_1, s_2)$, both moving actual coefficients. (An earlier formulation froze $a_{8,16} = 1$ and included the constant term as an unknown; the constant is killed by A_0' and contributes only a trivial direction.)

Remark 8.1 (Free parameter). §5 leaves $a_{8,16}$ free from the start. The E_2/E_3 analysis therefore applies to every slice, including $a_{8,16} = 1$, without a superset argument.

8.1 Explicit form of the seven conditions

Expanding (10) symbolically in (t_1, t_2, s_1, s_2) gives:

Lemma 8.2. *Each of the seven conditions has the form*

$$b_i(t_1, t_2) + s_1 \cdot M_i(t_1, t_2) + s_2 \cdot L_i(t_1, t_2) = 0, \qquad i = 1, \dots, 7, \tag{11}$$

where b_i is homogeneous cubic and M_i, L_i are linear in (t_1, t_2). Both s_1 and s_2 appear genuinely.

Proof. Exact expansion over K_5 in python-flint (script `exact_obstruction_K5.py`) of $W \cdot \mathrm{rhs}(t, s)$ with the homogeneous E_3 parametrization $x(t, s) = t_1^2 x_{11} + t_1 t_2 x_{12} + t_2^2 x_{22} + s_1 w_1 + s_2 w_2$ substituted; the SymPy expansion over $\mathbb{F}_{101}$ (script `C5_E2_obstruction_corrected.py`) is a cross-check. □

8.2 Elimination of (s_1, s_2)

For fixed $t \neq 0$, the seven equations are affine-linear in (s_1, s_2). A solution can exist only if the 7×3 matrix $[b(t) \mid M(t) \mid L(t)]$ has rank at most 2, i.e. only if all 3×3 minors vanish:

$$\det \begin{pmatrix} b_i & M_i & L_i \\ b_j & M_j & L_j \\ b_k & M_k & L_k \end{pmatrix} = 0, \qquad 1 \le i < j < k \le 7. \tag{12}$$

There are $\binom{7}{3} = 35$ such minors, each a homogeneous binary quintic in (t_1, t_2) (cubic × linear × linear).

Theorem 8.3 (No nonzero direction)**.** *The 35 quintics (12) span the degree-five homogeneous component $K_5[t_1, t_2]_5$ and generate the ideal $(t_1, t_2)^5$, i.e. they have Gröbner basis*

$$(t_1, t_2)^5 = \{ t_1^5,\ t_1^4 t_2,\ t_1^3 t_2^2,\ t_1^2 t_2^3,\ t_1 t_2^4,\ t_2^5 \} \tag{13}$$

exactly over K_5 (the 35×6 coefficient matrix has rank 6, spanning all binary quintics). Hence their common projective zero locus is empty and their only common affine zero in $\overline{K_5}^2$ is $(0, 0)$: there is no *nonzero t admitting (s_1, s_2) satisfying the seven E_2 conditions. The mod-101 Gröbner computation (lex order) is an independent cross-check.*

Proof. Exact computation over K_5: the 35 minors are homogeneous quintics, and their 35×6 coefficient matrix has rank 6, so

$$\mathrm{span}_{K_5}\{M_{ijk}\} = K_5[t_1, t_2]_5,$$

the full six-dimensional space of binary quintics. Hence every degree-five monomial lies in the ideal I they generate, i.e. $(t_1, t_2)^5 \subseteq I$; conversely each generator lies in $(t_1, t_2)^5$, so $I = (t_1, t_2)^5$. The radical is (t_1, t_2), so the affine variety over the algebraic closure is $\{(0, 0)\}$. Script: `exact_obstruction_K5.py`, which also writes the 35×6 matrix and a 6×6 nonzero minor to `logs/k5_minor_certificate.json` as a machine-readable certificate. The SymPy Gröbner basis over $GF(101)$ (script `C5_E2_obstruction_corrected.py`, saved, rerunnable; log archived) is a cross-check. □

Remark 8.4 (Interpretation). The top layer admits five characteristic-zero orbits, but none extends to a full solution: the E_4 kernel gives a two-dimensional space of directions $t \in K^2$, the E_3 layer imposes no further restriction, and then the seven E_2 compatibility conditions force $t = 0$ via the 35 quintic minors. The remaining case $t = 0$ is ruled out separately (Corollary 8.6). In other words, the top-layer solutions exist in isolation, but none extends to a Newton-normal-form pair with $[P, Q] = \lambda x^2$.

8.3 The case $t = 0$

Lemma 8.5 (Collapse at $t = 0$). *At $t = (0, 0)$ the E_2 layer equation reduces to*

$$2A_2B_0' = 0.$$

Proof. Because the target bracket identity $[P, Q] = x^2 = u^2y^{-4}$ contains only a y^{-4} component, the layer equation E_4 at y^{-3} is homogeneous:

$$2(A_2B_2' - A_2'B_2) + (A_1B_3' - 3A_1'B_3) = 0.$$

Consequently $t = (0, 0)$ forces $(A_1, B_2) \equiv (0, 0)$. In layer E_3 the quadratic driving term $K(t)$ vanishes identically at $t = 0$, leaving only the homogeneous kernel. In layer E_2 the terms $(A_1B_1' - A_1'B_1) - 2A_0'B_2$ then vanish identically, leaving $2A_2B_0' = 0$. □

Corollary 8.6 ($t = 0$ impossible). *By Lemma 8.5, at $t = 0$ we have $2A_2B_0' = 0$. Since $A_2 = u\alpha \neq 0$, this requires $B_0' = 0$, forcing $B_0(u)$ to be constant. Hence $b_{12,24} = 0$; in particular, $(12, 24)$ cannot be a vertex of $N(Q)$.*

Remark 8.7. The Lean formalization `e2_t0` (listing 20) proves $b_{12,24} = 0$ via a *different route*: a left-inverse certificate on the reduced E_2 equations with $t = 0$ supplied as hypotheses, rather than the $2A_2B_0' = 0$ argument above. Both routes reach the same conclusion.

*Listing 20: Mechanically extracted from **`Jacobian/Descent/E2/T0.lean`** (sha256 **`3559bcbdbff42d3c`**); statement verbatim, proof elided.*

```
theorem e2_t0 {L : Type*} [Field L] [CharZero L]
    (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0)
    (b_11_20 b_12_22 b_11_21 b_12_23 b_1_2 b_2_4 b_3_6 b_4_8 b_5_10 b_6_12 b_7_14
     ↪  b_8_16 b_9_18 b_10_20 b_11_22 b_12_24 : L)
    -- 19 reduced E2 equations (r2_1_1) ... (r2_19_37), hypotheses elided for brevity
    (ht1 : b_11_20 = 0)
    (ht2 : b_12_22 = 0) :
    b_12_24 = 0 := by
  -- proof body elided (see artifact)
```

8.4 Proof of the main theorem

Proof of Theorem 1.1. By Proposition 4.1, the layers satisfy $E_5, \dots, E_1$. Since $a_{1,0} \cdot a_{8,14} \cdot b_{2,1} \cdot b_{12,21} \neq 0$, Proposition 6.1 puts (A_2, B_3) in the torus orbit of the K_5 point for some root w of R. By torus transport (Listing 6) we may take it to be that point. Propositions 7.1–7.3 and Theorem 8.3 force $t = 0$. At $t = 0$, Lemma 8.5 and Corollary 8.6 give $B_0' = 0$, so $b_{12,24} = 0$

(in Lean, `e2_t0`, Listing 20). The last sentence of the theorem follows because NewtonNF2 requires $b_{12,24} \neq 0$. □

Proof of Corollary 1.2. By the proof of Theorem 1.1 we may take the top layer to be the K_5 point, where E_4, E_3, E_2 force $t = 0$; torus transport multiplies each coefficient by a nonzero scalar. At $t = 0$ we have $A_1 = B_2 = 0$ (Proposition 7.1). By Proposition 7.2, the coefficients of A_0 other than $a_{0,0}$ and the coefficients of B_1 are K_5-linear forms in $s = (b_{11,21}, b_{12,23})$, the two free unknowns of E_3. In particular $a_{8,16} = \kappa\, b_{12,23}$ with $\kappa \in K_5^{\times}$, and $b_{11,21}$ occurs in $a_{1,2}$ with a nonzero coefficient. Since $A_1 = 0$, the layer E_1, $A_1 B_0' - A_0' B_1 = 0$, reduces to $A_0' B_1 = 0$ in $K[u]$. If $B_1 = 0$, then $s = 0$. If $A_0' = 0$, then $a_{1,2} = a_{8,16} = 0$, so again $s = 0$. Hence $A_0 = a_{0,0}$ and $B_1 = 0$; in particular $a_{8,16} = 0$. Finally B_0 is constant by Corollary 8.6. In Lean (`lower_edge_rigidity`, generated by `lean/certgen/gen_a816.py`), the E_1 step uses the coefficients of $x^{17}y^{34}$, $x^{18}y^{36}$ and $x^{19}y^{38}$ in E_1 instead. At $t = 0$ they are quadratic forms in s that span all three quadratic monomials, and explicit K_5 multipliers give $b_{12,23}^2 = 0$ and $b_{11,21}^2 = 0$. The last one alone gives $a_{8,16} = 0$: it is $12\, a_{8,15} b_{12,24} - 8\, a_{8,16} b_{12,23} = -8\kappa\, b_{12,23}^2$.
A second proof of $a_{8,16} = 0$, outside Lean and independent of the descent, is an explicit certificate. At the K_5 point, let $e_1, \dots, e_{75}$ be the coefficients of $[P, Q] - x^2$ in the layers $E_4, \dots, E_1$, as built by `scripts/a816_full.sing`; their unknowns are the 51 lower coefficients of P and Q that occur. There is an explicit identity

$$a_{8,16}^2 = \sum_{k=1}^{75} H_k\, e_k, \qquad H_k \in K_5[a, b],$$

stored in Rabinowitsch form, $1 = \sum_k z^2 H_k\, e_k - (1 + a_{8,16} z)(a_{8,16} z - 1)$, as the 76 cofactors of `scripts/a816_certificate/a816_lift.txt` (3464 terms, coefficients of at most 493 digits). It is built from the layer structure. Give the coefficients of A_1, B_2 depth 1, those of A_0, B_1 depth 2 and those of B_0 depth 3; each equation of E_{5-d} is then homogeneous of depth d and linear in the unknowns of depth d. Row reduction of these linear parts in E_4, E_3, E_2 (ranks 17, 18, 12 over K_5) expresses every unknown, modulo the layer ideal, as a polynomial in four free ones, $\tau = (b_{11,20}, b_{12,22})$ of depth 1 and $\sigma = (a_{8,16}, b_{11,21})$ of depth 2. The 32 products of depth 4 formed from the reduced generators span all 14 monomials of depth 4 in $K_5[\tau, \sigma]$, and lifting back gives the H_k. The identity is checked exactly by SINGULAR over $\mathbb{Q}(w)$ and independently by python-flint, which rebuilds $[P, Q]$ from P and Q (`scripts/a816_certificate/verify_bundle.sh`). Hence $a_{8,16} = 0$ at the K_5 point once E_1 is included. Torus transport multiplies $a_{8,16}$ by a nonzero scalar, so $a_{8,16} = 0$ on the whole orbit. □

Remark 8.8 (Further rigidity, outside the certificate). The same span statement puts every monomial of depth at least 4 in τ, σ into the layer ideal, so each of the 51 unknowns (the non-constant lower coefficients of P and Q) is nilpotent modulo it: at the K_5 point the only solution of $E_4, \dots, E_1$ is $P = P_2 +$ const, $Q = Q_3 +$ const. Explicitly, let x be an unknown of depth d, and let $p_x \in K_5[\tau, \sigma]$ be the polynomial that the row reduction gives, with $x \equiv p_x$ modulo the layer ideal ($p_x = x$ for a free unknown). It is homogeneous of depth d with no constant term. So $x^{\lceil 4/d \rceil} \equiv p_x^{\lceil 4/d \rceil}$, and every monomial of $p_x^{\lceil 4/d \rceil}$ is divisible by a monomial of depth exactly 4. The certificates behind this are written out and checked exactly by two independent programs (`scripts/a816_rigidity/`): the 47 identities $x - p_x = \sum_k D_{x,k} e_k$ for the pivot unknowns, and the 14 identities $m = \sum_k H_{m,k} e_k$ for the monomials m of depth 4. The zero-set form of this statement, $P = P_2 +$ const and $Q = Q_3 +$ const, is the Lean theorem

`lower_edge_rigidity` (Corollary 1.2); the nilpotency modulo the layer ideal is not part of the Lean formalization. Neither is used elsewhere.

Listing 21: Mechanically extracted from ***Jacobian/A816/Final.lean*** *(sha256* ***b7f3ed6341617a44****); statement verbatim, proof elided.*

```
theorem lower_edge_rigidity {L : Type*} [Field L] [CharZero L] (P Q : MvPolynomial
 ↪  (Fin 2) L) (lam : L)
    (hP : forall m ∈ P.support, inNP m) (hQ : forall m ∈ Q.support, inNQ m)
    (v10 : P.coeff (mono 1 0) ≠ 0) (v814 : P.coeff (mono 8 14) ≠ 0)
    (w21 : Q.coeff (mono 2 1) ≠ 0) (w1221 : Q.coeff (mono 12 21) ≠ 0)
    (hJ : jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2) :
    (forall i j, 2 * i ≤ j + 1 → (i, j) ≠ (0, 0) → P.coeff (mono i j) = 0) /\
      (forall k l, 2 * k ≤ l + 2 → (k, l) ≠ (0, 0) → Q.coeff (mono k l) = 0) := by
  -- proof body elided (see artifact)
```

Listing 22: Mechanically extracted from ***Jacobian/A816/Final.lean*** *(sha256* ***b7f3ed6341617a44****); statement verbatim, proof elided.*

```
theorem main_theorem_lower_edge {L : Type*} [Field L] [CharZero L] :
    ¬ exists (P Q : MvPolynomial (Fin 2) L) (lam : L), lam ≠ 0 /\ NewtonNF2 P Q /\
     ↪  jac P Q = MvPolynomial.C lam * MvPolynomial.X 0 ^ 2 := by
  -- proof body elided (see artifact)
```

Proof of Corollary 1.4. Proposition 4.3 of [1], then Theorem 1.1. □

Listing 23: Mechanically extracted from ***Jacobian/BranchAbFinal.lean*** *(sha256* ***cd4e207adba5cb7b****); statement verbatim, proof elided.*

```
theorem no_completion_K5 {L : Type*} [Field L] [CharZero L] (w : L)
    (hw : w ^ 5 - w ^ 4 + 3 * w ^ 3 + 3 * w ^ 2 + 26 = 0) (lam : L)
    (A₀ A₁ A₂ B₀ B₁ B₂ B₃ : Polynomial L) (p₀ p₁ p₂ q₀ q₁ q₂ q₃ : MvPolynomial (Fin
     ↪  2) L)
    (hp₀ : X 1 ^ 0 * p₀ = evH A₀) (hp₁ : X 1 ^ 1 * p₁ = evH A₁) (hp₂ : X 1 ^ 2 * p₂ =
     ↪  evH A₂)
    (hq₀ : X 1 ^ 0 * q₀ = evH B₀) (hq₁ : X 1 ^ 1 * q₁ = evH B₁) (hq₂ : X 1 ^ 2 * q₂ =
     ↪  evH B₂)
    (hq₃ : X 1 ^ 3 * q₃ = evH B₃)
    (hJ : jac (p₀ + p₁ + p₂) (q₀ + q₁ + q₂ + q₃) = C lam * X 0 ^ 2)
    (sA₀ : forall i, A₀.coeff i ≠ 0 → 0 ≤ i /\ i ≤ 8)
    (sA₁ : forall i, A₁.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8)
    (sA₂ : forall i, A₂.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 8)
    (sB₀ : forall i, B₀.coeff i ≠ 0 → 0 ≤ i /\ i ≤ 12)
    (sB₁ : forall i, B₁.coeff i ≠ 0 → 1 ≤ i /\ i ≤ 12)
    (sB₂ : forall i, B₂.coeff i ≠ 0 → 2 ≤ i /\ i ≤ 12)
    (sB₃ : forall i, B₃.coeff i ≠ 0 → 2 ≤ i /\ i ≤ 12)
    (ρ σ ε : L) (hρ : ρ ≠ 0) (hσ : σ ≠ 0) (hε : ε ≠ 0)
    (t_a_1_0 : (A₂.coeff 1) = ρ * ε ^ (0 : Nat) * ((1 : L)))
    (t_a_2_2 : (A₂.coeff 2) = ρ * ε ^ (1 : Nat) * (((303) / 16 : L) + ((879) / 128 :
     ↪  L) * w + ((-389) / 128 : L) * w ^ (2 : Nat) + ((-51) / 128 : L) * w ^ (3 :
     ↪  Nat) + ((49) / 128 : L) * w ^ (4 : Nat)))
    (t_a_3_4 : (A₂.coeff 3) = ρ * ε ^ (2 : Nat) * (((315) / 16 : L) + ((693) / 8 : L)
     ↪  * w + ((-147) / 8 : L) * w ^ (2 : Nat) + ((-189) / 8 : L) * w ^ (3 : Nat) +
     ↪  ((147) / 16 : L) * w ^ (4 : Nat)))
```

```
  (t_a_4_6 : (A₂.coeff 4) = ρ * ε ^ (3 : Nat) * (((-79653) / 64 : L) + ((29463) /
   ↪  64 : L) * w + ((7371) / 128 : L) * w ^ (2 : Nat) + ((-4767) / 16 : L) * w ^
   ↪  (3 : Nat) + ((8001) / 128 : L) * w ^ (4 : Nat)))
  (t_a_5_8 : (A₂.coeff 5) = ρ * ε ^ (4 : Nat) * (((-33519) / 4 : L) + ((-6843) / 4
   ↪  : L) * w + ((8181) / 8 : L) * w ^ (2 : Nat) + ((-1713) / 2 : L) * w ^ (3 :
   ↪  Nat) + ((-1305) / 8 : L) * w ^ (4 : Nat)))
  (t_a_6_10 : (A₂.coeff 6) = ρ * ε ^ (5 : Nat) * (((-8805) / 8 : L) + ((-76965) / 8
   ↪  : L) * w + ((11523) / 16 : L) * w ^ (2 : Nat) + ((6927) / 4 : L) * w ^ (3 :
   ↪  Nat) + ((-18579) / 16 : L) * w ^ (4 : Nat)))
  (t_a_7_12 : (A₂.coeff 7) = ρ * ε ^ (6 : Nat) * (((389740) / 17 : L) + ((-51922) /
   ↪  17 : L) * w + ((-48572) / 17 : L) * w ^ (2 : Nat) + ((63239) / 17 : L) * w ^
   ↪  (3 : Nat) + ((-6475) / 17 : L) * w ^ (4 : Nat)))
  (t_a_8_14 : (A₂.coeff 8) = ρ * ε ^ (7 : Nat) * (((476437) / 68 : L) + ((314405) /
   ↪  68 : L) * w + ((-174571) / 136 : L) * w ^ (2 : Nat) + ((1574) / 17 : L) * w ^
   ↪  (3 : Nat) + ((82783) / 136 : L) * w ^ (4 : Nat)))
  (t_b_2_1 : (B₃.coeff 2) = σ * ε ^ (0 : Nat) * ((1 : L)))
  (t_b_3_3 : (B₃.coeff 3) = σ * ε ^ (1 : Nat) * (((101) / 8 : L) + ((293) / 64 : L)
   ↪  * w + ((-389) / 192 : L) * w ^ (2 : Nat) + ((-17) / 64 : L) * w ^ (3 : Nat) +
   ↪  ((49) / 192 : L) * w ^ (4 : Nat)))
  (t_b_4_5 : (B₃.coeff 4) = σ * ε ^ (2 : Nat) * (((315) / 16 : L) + ((693) / 8 : L)
   ↪  * w + ((-147) / 8 : L) * w ^ (2 : Nat) + ((-189) / 8 : L) * w ^ (3 : Nat) +
   ↪  ((147) / 16 : L) * w ^ (4 : Nat)))
  (t_b_5_7 : (B₃.coeff 5) = σ * ε ^ (3 : Nat) * (((-11379) / 8 : L) + ((4209) / 8 :
   ↪  L) * w + ((1053) / 16 : L) * w ^ (2 : Nat) + ((-681) / 2 : L) * w ^ (3 : Nat)
   ↪  + ((1143) / 16 : L) * w ^ (4 : Nat)))
  (t_b_6_9 : (B₃.coeff 6) = σ * ε ^ (4 : Nat) * (((-695299) / 48 : L) + ((-48367) /
   ↪  48 : L) * w + ((52679) / 32 : L) * w ^ (2 : Nat) + ((-45661) / 24 : L) * w ^
   ↪  (3 : Nat) + ((-2211) / 32 : L) * w ^ (4 : Nat)))
  (t_b_7_11 : (B₃.coeff 7) = σ * ε ^ (5 : Nat) * (((-309895) / 8 : L) + ((-251275)
   ↪  / 8 : L) * w + ((101481) / 16 : L) * w ^ (2 : Nat) + ((1811) / 4 : L) * w ^
   ↪  (3 : Nat) + ((-60489) / 16 : L) * w ^ (4 : Nat)))
  (t_b_8_13 : (B₃.coeff 8) = σ * ε ^ (6 : Nat) * (((1733348) / 13 : L) +
   ↪  ((-2368725) / 26 : L) * w + ((-184976) / 13 : L) * w ^ (2 : Nat) + ((1819939)
   ↪  / 52 : L) * w ^ (3 : Nat) + ((-569071) / 52 : L) * w ^ (4 : Nat)))
  (t_b_9_15 : (B₃.coeff 9) = σ * ε ^ (7 : Nat) * (((18736247) / 39 : L) +
   ↪  ((1087133) / 13 : L) * w + ((-5656777) / 78 : L) * w ^ (2 : Nat) +
   ↪  ((1993384) / 39 : L) * w ^ (3 : Nat) + ((971365) / 78 : L) * w ^ (4 : Nat)))
  (t_b_10_17 : (B₃.coeff 10) = σ * ε ^ (8 : Nat) * (((305098) / 17 : L) +
   ↪  ((5784066) / 17 : L) * w + ((-308335) / 17 : L) * w ^ (2 : Nat) + ((-950608)
   ↪  / 17 : L) * w ^ (3 : Nat) + ((754021) / 17 : L) * w ^ (4 : Nat)))
  (t_b_11_19 : (B₃.coeff 11) = σ * ε ^ (9 : Nat) * (((-149721182) / 323 : L) +
   ↪  ((26391690) / 323 : L) * w + ((21560033) / 323 : L) * w ^ (2 : Nat) +
   ↪  ((-24378412) / 323 : L) * w ^ (3 : Nat) + ((2786935) / 323 : L) * w ^ (4 :
   ↪  Nat)))
  (t_b_12_21 : (B₃.coeff 12) = σ * ε ^ (10 : Nat) * (((-91912444) / 969 : L) +
   ↪  ((-20715500) / 323 : L) * w + ((17474146) / 969 : L) * w ^ (2 : Nat) +
   ↪  ((-1623428) / 969 : L) * w ^ (3 : Nat) + ((-8783302) / 969 : L) * w ^ (4 :
   ↪  Nat)))
  (hv : B₀.coeff 12 ≠ 0) : False := by
-- proof body elided (see artifact)
```

9 Exact characteristic-0 descent

The proof is exact over K_5 (items (i)–(ii) below).

Lemma 9.1 (Base change for the K_5 computations). *$R(w) = w^5 - w^4 + 3w^3 + 3w^2 + 26$ is irreducible and separable over $\mathbb{Q}$, so $K_5 = \mathbb{Q}[w]/(R)$ is a number field. The coefficients of the E_4, E_3, E_2 matrices and of the seven compatibility polynomials lie in K_5. Exact rank identities and exact ideal identities over K_5 therefore remain valid under every characteristic-zero field extension.*

Proof. R is irreducible modulo 67 (a single quintic factor), hence irreducible over $\mathbb{Q}$, and separable since its discriminant $784{,}822{,}272 = 2^{12} \cdot 3 \cdot 13 \cdot 17^3$ is nonzero. Hence K_5 is a field. Rank of a matrix and equality of ideals are preserved under field extension: rank is the size of the largest nonzero minor, and ideal membership is witnessed by explicit polynomial combinations, both of which persist after extending scalars. □

Lemma 9.2 (Exact obstruction). *Let $\rho \in \overline{\mathbb{Q}}$ be any root of W, and put $K = \mathbb{Q}(\rho)$. The E_2 obstruction—no nonzero (t_1, t_2) satisfies the seven compatibility conditions—holds over K.*

Proof. By Lemma 9.1, the exact rank and ideal identities over K_5 persist under every characteristic-zero field extension. Since $K_5 \cong \mathbb{Q}(\rho^7)$ for every root ρ of W (the $S = a_6^7$ parametrization; Remark 9.3), the obstruction holds over $\mathbb{Q}(\rho)$, hence over every characteristic-zero field containing such a root. The ingredients are the exact characteristic-0 computations (scripts `exact_ranks_K5.py`, `exact_obstruction_K5.py`):

(i) *Exact characteristic-0 ranks.* The ranks are computed exactly over the quintic field $K_5 = \mathbb{Q}[w]/(w^5 - w^4 + 3w^3 + 3w^2 + 26)$ (script `exact_ranks_K5.py`): $E_4 = 17$, $E_3 = 18$, $E_2 = 12$.

(ii) *Exact descent over K_5.* The entire $E_4 \to E_3 \to E_2 \to 35$-minor pipeline is redone exactly over K_5 (script `exact_obstruction_K5.py`, ~1 s): E_3 is solvable for every t, the left nullity of E_2 is 7, and the 35 minors span *all* binary quintics (the 35×6 coefficient matrix has rank 6 over K_5). Hence the only common zero is $t = 0$. End-to-end residuals against the original bracket equations vanish exactly. The $t = 0$ case is excluded by the vertex $(12, 24)$ argument, which is valid in every characteristic other than 2 and 3 (it uses $2 \neq 0$ and $12 \neq 0$; see §8.3). The Lean formalization (`no_completion_K5`, listing 23, via `main_theorem_of_chart`, listing 7) quantifies over all roots of R and all torus parameters, covering all top-layer solutions by Proposition 6.1, whose $m = 7$ classification is formalized (`chartClassification_holds`, §6.4).

□

Remark 9.3 (On the derivation of W). The polynomial W was originally imported from the reference data (see `W_IMPORT_DISCLOSURE.md`). It is now *independently derived*: the minimal polynomial of a_6^7, computed from the exactly verified top-layer point over K_5, equals the quintic V (hence $W(a_6) = V(a_6^7)$) coefficient for coefficient (script `compare_V.gp`). The import caveat in `W_IMPORT_DISCLOSURE.md` is retired for W itself. The mod-101 reduction matches the independently computed elimination up to scalar 54 as a cross-check.

10 Conclusion

We have eliminated branches (a) and (b) of the GGHV $(72, 108)$ candidate—case (2) of [1, Proposition 4.3]—by a computational audit:

- source-gated base objects (polygons, bracket, and φ verified against GGHV v1);
- exact layer reduction (92 equations, symbolic identity);
- top-layer determination (degree-35 eliminant, one real solution);
- descent with explicit ranks and unconditional E_3 solvability;
- minor-span obstruction ($(t_1, t_2)^5$) closing the E_2 conditions;
- exact K_5-descent (Lean `no_completion_K5` via `main_theorem_of_chart`) covering all roots of R and all torus parameters;

- an algebraic proof of the $m = 7$ top-layer classification (Proposition 6.1 in the normalized chart), using reversed coordinates, six orbit relations and a Sylvester resultant (§6.4). It is formalized in Lean, so the NewtonNF2 form of Theorem 1.1 is the kernel-checked theorem `main_theorem`, independent of GGHV Proposition 4.3;
- lower-edge rigidity (Corollary 1.2): $P = P_2 + a_{0,0}$ and $Q = Q_3 + b_{0,0}$, kernel-checked as `lower_edge_rigidity`, so that the vertex $(8, 16)$ refutes NewtonNF2 as well (`main_theorem_lower_edge`).

11 Prior work, attribution, and independence

Prior work. F. Santibañez-Leal's CAOS records EXP-061 and EXP-065 treat exactly these cases with sampled certificates, later downgraded by his own audit. We cite them here for completeness.

Attribution. The layer reduction strategy ($u = xy^2$, five triangular layers), the identification of the Davenport–Stothers top layer, the descent architecture, and the modular certificate approach originate in the initial computational package (`branch_ab_elimination.zip`, MD5 `0cfa49d0`…). Its full-formulation certificate at $p = 10^9 + 9$ and $p = 10^9 + 21$, with lifting argument, is a second, independent route to the same obstruction (see the Appendix). This work reimplements the reduction independently in SINGULAR (C1) and recomputes the descent, but the overall strategy is not original to us.

Characteristic-0 core. The characteristic-zero computations—the Belyi counts (`belyi_counts_m357.py`), the exact top-layer point over K_5 (`e5_exact_K5.json`), the derivation of W (`compare_V.gp`), the exact ranks (`exact_ranks_K5.py`), the exact $E_4 \to E_3 \to E_2$ descent (`exact_obstruction_K5.py`), and the msolve-parametrization check (`verify_msolve_param.py`)—are documented in the scripts (see their attribution headers). Tools: python-flint 0.9.0 (exact K_5 arithmetic), sympy 1.14.0, PARI/GP 2.15.4, Singular 4.3.2. The independence case rests on the artifacts, not on who ran them: the Lean kernel checks every formalized proof; the Singular Gröbner computations (`a816_full.sing`, `chart_system_Q.sing`) can be rerun from the archive; the $a_{8,16}$ certificate (`scripts/a816_certificate/`) is checked by two independent programs; and the Belyi counts were verified by an independent Murnaghan–Nakayama implementation.

Independent AI-assisted computational audit (completed 2026-09-25). An independent AI-assisted computational audit reached the same conclusion by an independent route without access to the present scripts. The auditing agent received only a short brief—the degree pair $(72, 108)$ and the instruction to work from GGHV Proposition 4.3—plus GGHV v1 itself. It independently produced a proof using weight grading, a Belyi-map count (at most 35 complex points via Riemann's existence theorem and the Frobenius character formula), exact modular certificates at $p = 32003$ and $p = 536870909$ (Singular, Macaulay2, msolve), and Hensel lifting, reaching the same nonexistence conclusion and the stronger lower-edge rigidity $a_{8,16} = b_{12,24} = 0$. Materials: `tier1_blind_gghv43_nf2/` [8], committed as `37c73b67` on 2026-09-25. The evidential weight rests on the reproduced artifacts and computations, not on the identity of the auditing system.

Remark 11.1 (What remains). The application of Theorem 1.1 to the Jacobian conjecture, Corollary 1.4, is conditional on [1, Proposition 4.3] (Remark 1.5). The remaining open case is branch (c) (case (1) of Proposition 2.1), whose Newton polygons share the same top-layer E_5 equation and are amenable to the same machinery with additional lower layers. The status of the $(72, 108)$ candidate as a whole remains underdetermined pending branch (c).

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[7] B. D. McCrary, Muse, *The* $(72, 108)$ *candidate for the planar Jacobian conjecture: a containment note*, Zenodo doi:10.5281/zenodo.22941057 (2026).

[8] Claude (Anthropic), *Independent blind replication: GGHV Proposition 4.3, normal form (2)*, `tier1_blind_gghv43_nf2/` in B. D. McCrary, *2D-Jacobian-Conjecture-Workspace*, GitHub (2026), commit `37c73b67` (2026-09-25).

A Computational archive

All scripts used in this audit are archived in `tier2_branch_ab_scripts.zip`:

- `C1_layer_reduction.sing` — exact 92-equation layer reduction (symbolic identity);
- `source_gate_verify.py` — asserts GGHV v1 lines 33, 492–495, 596–600;
- E_4 (an 18×19 matrix of rank 17), kernel 2 — verified within `C5_E2_obstruction_corrected.py` (with $A_2 = u\alpha$, $B_3 = u^2\beta$);
- `C5_E2_obstruction_corrected.py` — E_3 (a 19×20 matrix of rank 18) and $v^T K \equiv 0$ (with residual asserts);
- Top-layer elimination, SINGULAR factorization over $\mathbb{Q}$, discriminant and Sturm computations;
- E_2 rank verification (injectivity of the 19×12 operator).

The generator of the Lean proof of Theorem 6.5 (§6.4) ships with the Lean project, in `lean/certgen/chartproof/`. It finds the orbit relations by exact interpolation, computes the certificates by weighted-homogeneous linear algebra over $\mathbb{Q}$, and verifies all 105 step identities in SymPy; `regenerate_chartproof.sh` checks byte-identical regeneration. The E_2 seven-condition expansion ($b_i + s_1 M_i + s_2 L_i = 0$) and the 35-minor Gröbner basis over $\mathbb{F}_{101}$ (§8) are computed in the saved, rerunnable script `C5_E2_obstruction_corrected.py`; the log `C5_E2_obstruction_corrected.log` is archived. Modular computations use $p = 101$; characteristic-zero claims (irreducibility over $\mathbb{Q}$, discriminant nonzero, exactly one real root) are over $\mathbb{Q}$. The earlier E_4/E_3 scripts using unscaled (α, β) are superseded and excluded from the archive.

Reproducibility

All artifacts are archived in the public repository, directory `branch_ab_v19/`; the directory `JC2D-branches_a_b_c_unified/branches_a_b/` of the same repository is an identical copy. The artifacts of the first v19 release are at commit `759770093f853907d13d4d492430f71f9e738e41`. The revisions of October 5–7, 2026 add the files listed below. Apart from those additions and the lines that wire them in (in `lean/Jacobian.lean`, `lean/AxiomsAudit.lean`, `lean/verify_branch_ab_lean.sh` and the two regeneration scripts of `lean/certgen/`), the Lean project sources are byte-identical to the v18 sources. The full build of those sources is recorded in `BUILD_STATUS.md`, and so is the build of the new Lean files. `#print axioms` on `main_theorem` reports only `propext`, `Classical.choice`, `Quot.sound`. The certificate chain is generator → certificate → Lean theorem: `lean/certgen/chartproof/` generates the ChartProof modules (byte-identical regeneration checked by `regenerate_chartproof.sh`), `independent_check.py` re-verifies all 111 identities over $\mathbb{Z}$ independently of the generator, and the Lean kernel checks every proof term. The K_5 descent scripts (`exact_ranks_K5.py`, `exact_obstruction_K5.py`) run from `lean/certgen/` with python-flint 0.9.0; `logs/v19_scripts.log` records their exact output and exit codes, and `logs/k5_minor_certificate.json` carries the machine-readable 35×6 minor matrix with its rank certificate. The additions of that revision are:

- the $a_{8,16}$ certificate, `scripts/a816_certificate/`;
- the Lean files `lean/Jacobian/B26.lean`, `lean/Jacobian/B26/`, `lean/Jacobian/B26Count.lean` and `lean/Jacobian/B26Irred.lean`, with their generator, certificates and statement check in `scripts/b26_m5_eliminant/`;
- the c-recursion validation, `scripts/b22_structured/`;
- the certificates of Remark 8.8, `scripts/a816_rigidity/`;
- the Lean proof of Corollary 1.2, `lean/Jacobian/A816/`, with its generator `lean/certgen/gen_a816.py`, its statement check `lean/certgen/check_a816_statement.py` and its controls `lean/controls_a816.sh`.

`verify_v19.sh` runs the checks of the first three; `scripts/a816_rigidity/run.sh` runs the fourth (about 8 min); `lean/verify_branch_ab_lean.sh`, `lean/controls_a816.sh` and `lean/certgen/check_regeneration.sh` check the fifth.

B Original certificate (second route)

Attribution. The following certificate was computed by Claude (Anthropic) during review, as part of the original package `branch_ab_elimination.zip`. It is included here as a second, independent route to the same obstruction.

Method. Full-formulation (remainder-based) certificate: constructs the E_2 conditions via polynomial remainder rather than left-null-space, uses FLINT for linear algebra (rather than hand-coded Gaussian elimination), and runs at $p = 10^9 + 9$ and $p = 10^9 + 21$ (rather than $p = 101$).

Shared inputs. Both routes share: GGHV's normal form (Proposition 4.3), the exact layer reduction formula ($u = xy^2$, five triangular layers), and the eliminating polynomial W.

Result. At $p = 1000000009$: 7 simple roots of W in $\mathbb{F}_p$, 2 certified with `E4_rank=17`, `E3_rank=18`, `E3_consistent_at_nodes=True`. At $p = 1000000021$: 2 simple roots, 2 certified. All show `certified=True`.

Coverage of all top-layer solutions. The Lean theorems `no_completion_K5` (listing 23) and `main_theorem_of_chart` (listing 7) quantify over all roots w of R and all torus parameters $\rho, \sigma, \varepsilon \neq 0$, so "no top-layer solution admits a completion" holds for every solution of the top-layer system by Proposition 6.1 (for $m = 7$ formalized as `chartClassification_holds`), not just the K_5-point. The exact K_5 computation (scripts `exact_ranks_K5.py`, `exact_obstruction_K5.py`) is an independent cross-check of the same statement.

Scripts: `cert_branch_ab.py`, `cert_setup.py`; log: `cert_branch_ab.log`. All archived in `tier2_branch_ab_scripts.zip`.

C Computational and AI-assisted research disclosure

Two AI systems contributed to this work. Claude (Anthropic) served as pre-submission reviewer and computed the characteristic-zero verification scripts (`belyi_counts_m357.py`, `exact_ranks_K5.py`, `exact_obstruction_K5.py`, `verify_msolve_param.py`; see their attribution headers), the chart audit (`audit_chart.py`), the $a_{8,16} = 0$ Gröbner computation (`a816_full.sing`) and its explicit certificate over K_5 (`scripts/a816_certificate/`), the Lean formalization of the $m = 3$ and $m = 5$ classifications (`lean/Jacobian/B26.lean`) and of Corollary 1.2 (`lean/Jacobian/A816/`), the Murnaghan–Nakayama implementation (`frobenius.py`), and the independent Lean build. During the v17 review Claude also found the algebraic proof of the $m = 7$ top-layer classification in §6.4 (the reversed coordinates, the six orbit relations and their certificates) and wrote its Lean formalization (`lean/Jacobian/ChartProof`, with the generator `lean/certgen/chartproof`). Claude further performed the independent AI-assisted computational audit described above (see `tier1_blind_gghv43_nf2/` commit history). Muse (Meta; also called CAIC), the author's personal AI assistant, generated the Lean listings mechanically from committed source (`gen_listings.py`), computed the $m = 5$ eliminant and back-substitution formulas (`scripts/b26_m5_eliminant/`) and the c-recursion form of the chart system (`scripts/b22_structured/`), and assisted with manuscript drafting; all such outputs were verified by execution or by the Lean kernel. The author (B. D. McCrary) directed the investigation, verified the results, and takes responsibility for the manuscript. The mathematical argument stands on the artifacts above, not on these attributions.